

CLASS 9 PHYSICS FOUNDATION

Complete Exam-Ready Notes

Structured with Quick Summaries · Real-Life Examples · Formula Sheets
Derivations · Memory Hooks · Common Mistakes · Chapter-End Revision

Chapter 0: Basic Mathematics for Physics

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Comfort with algebraic expressions, powers and simple equations.
- Basic idea of a right-angled triangle and its sides.
- Familiarity with plotting points on a graph.
- Knowledge of ratios and proportion.

0.1 Algebra and Quadratic Equations in Physics

 **Estimated time:** 14 mins **Difficulty:**  Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Physics constantly requires rearranging formulae, solving quadratic equations and handling powers of ten, so fluency in these algebraic tools is the first requirement.

Real-Life Example

Finding the time at which a thrown ball hits the ground means solving a quadratic equation — physics gives the equation, algebra gives the answer.

Detailed Explanation

Definition

A quadratic equation is an equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$. Its two roots are given by the quadratic formula.

Key Idea

In physics the letters carry units, so an equation must be rearranged symbolically first and numbers substituted only at the very end. This avoids rounding errors and makes unit checking possible.

Working Principle

Whatever operation is applied to one side of an equation must be applied to the other. To make a particular symbol the subject, move every other term away from it by applying the inverse operation in reverse order.

Applications

- $s = ut + \frac{1}{2}at^2$ is a quadratic in t and is solved with the quadratic formula.
- Powers of ten make very large and very small physical quantities manageable.
- Binomial approximation $(1 + x)^n \approx 1 + nx$ for small x simplifies many physics expressions.

✓ **Points to Remember (Exam Focus)**

- ✓ Rearrange symbolically first, substitute numbers last.
- ✓ A quadratic in physics often has one physically meaningless root (e.g. negative time) — discard it.
- ✓ $a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$.
- ✓ Standard form: a number between 1 and 10 multiplied by a power of ten.
- ✓ Always check that both sides of an equation have the same units.

⚠ Common Student Mistakes

- ⚠ Substituting numbers too early and then being unable to check units.
- ⚠ Keeping both roots of a quadratic when one is physically impossible.
- ⚠ Cancelling terms across a plus sign instead of across a multiplication.



Memory Hook

'Symbols first, numbers last' — and 'if the units do not match, the equation is wrong'.



Advanced Insights

Dimensional analysis can reconstruct a formula up to a dimensionless constant. Knowing that the period of a pendulum can depend only on length and g forces $T \propto \sqrt{l/g}$ without solving a single equation of motion.

Formula Section



Quadratic Formula

Difficulty	● Medium ★★☆☆
Formula	$x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$
Meaning of Variables	a, b, c = coefficients of the quadratic $ax^2 + bx + c = 0$; x = the two roots.
When to Use	Whenever an equation of motion or energy leads to a squared unknown, e.g. finding time from $s = ut + \frac{1}{2}at^2$.
Common Mistakes	Forgetting the $\pm$ sign, or dividing only the square-root term by $2a$ instead of the whole numerator.
Shortcut / Memory Trick	If $b^2 - 4ac$ is negative, there is no real solution — check the physics, something is wrong.

0.2 Trigonometry and Vectors for Physics

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

Quick Summary

Trigonometric ratios resolve a quantity into components along two perpendicular directions, which is the standard way of handling forces and velocities at an angle.

Real-Life Example

When you pull a suitcase by its handle, only the horizontal part of your pull moves it forward — trigonometry tells you exactly how much.

Detailed Explanation

Definition

For a right-angled triangle, $\sin \theta = \text{opposite/hypotenuse}$, $\cos \theta = \text{adjacent/hypotenuse}$, $\tan \theta = \text{opposite/adjacent}$. Resolution of a vector is the splitting of a vector into two mutually perpendicular components.

Key Idea

A vector of magnitude A at angle θ to the x -axis has components $A \cos \theta$ along x and $A \sin \theta$ along y . The component along the direction of the angle always carries the cosine.

Working Principle

Components are added separately along each axis, and the resultant is recovered using Pythagoras: $R = \sqrt{R_x^2 + R_y^2}$, with direction $\tan \theta = R_y/R_x$.

Applications

- Resolving weight on an inclined plane into $mg \sin \theta$ along the slope and $mg \cos \theta$ perpendicular to it.
- Finding the horizontal and vertical components of a projectile's initial velocity.
- Adding two forces acting at an angle to find the single resultant force.

✓ Points to Remember (Exam Focus)

- ✓ $\sin 0^\circ = 0$, $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{1}{\sqrt{2}}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\sin 90^\circ = 1$.
- ✓ $\cos$ values run in the reverse order of $\sin$ values.
- ✓ $\sin^2 \theta + \cos^2 \theta = 1$ for every angle.
- ✓ The component ALONG the angle uses $\cos$; the component perpendicular uses $\sin$.
- ✓ Scalars have magnitude only; vectors have both magnitude and direction.

Common Student Mistakes

-  Interchanging $\sin$ and $\cos$ when resolving — always identify which side the angle is measured from.
-  Adding vector magnitudes arithmetically instead of resolving into components first.
-  Forgetting that the resultant of two equal and opposite vectors is zero, not double.

 Memory Hook

'SOH-CAH-TOA' — and 'the Cosine hugs the angle, the Sine stands opposite'.

 Advanced Insights

For small angles measured in radians, $\sin \theta \approx \tan \theta \approx \theta$ and $\cos \theta \approx 1$. This small-angle approximation is what makes a pendulum's motion simple harmonic and is used throughout optics and oscillation theory.

Formula Section **Resolution of a Vector**

Difficulty	● Medium ★★☆☆
Formula	$A_x = A \cos \theta$, $A_y = A \sin \theta$
Meaning of Variables	A = magnitude of the vector, θ = angle with the x-axis, A_x and A_y = components.
When to Use	Whenever a force, velocity or displacement acts at an angle and must be split along two perpendicular directions.
Common Mistakes	Using sin where cos is required because the angle was measured from the wrong axis.
Shortcut / Memory Trick	Draw the triangle first; the side touching the angle is the cosine component.

 Resultant of Perpendicular Vectors

Difficulty	● Easy ★☆☆
Formula	$R = \sqrt{R_x^2 + R_y^2}$
Meaning of Variables	R = magnitude of the resultant, R_x and R_y = perpendicular components.
When to Use	To recombine components into a single vector after adding them axis by axis.
Common Mistakes	Adding the components directly instead of squaring, adding, then taking the square root.
Shortcut / Memory Trick	This is just Pythagoras' theorem applied to vectors.

II Quick Recap — Topics Covered So Far

- Algebra and Quadratic Equations in Physics — Physics constantly requires rearranging formulae, solving quadratic equations and handling powers of ten, so fluency in these algebraic tools is the first requirement.

- Trigonometry and Vectors for Physics — Trigonometric ratios resolve a quantity into components along two perpendicular directions, which is the standard way of handling forces and velocities at an angle.

0.3 Graphs and Their Physical Meaning

Estimated time: 13 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

The slope of a graph gives the rate of change of the quantity plotted on the y-axis, and the area under a graph gives the product of the two plotted quantities.

Real-Life Example

A cyclist's speedometer trace tells the whole story: how steeply it rises gives the acceleration, and the area beneath it gives the distance travelled.

Detailed Explanation

Definition

Slope of a straight-line graph = (change in y) / (change in x). The area under a graph between two points represents the product of the y-quantity and the x-quantity over that interval.

Key Idea

Two operations extract all the physics from a graph. Slope always answers 'how fast is y changing with x' and area always answers 'what is the accumulated product'. Everything else is interpretation.

Working Principle

On a distance–time graph the slope is speed. On a velocity–time graph the slope is acceleration and the area is displacement. On a force–displacement graph the area is work done.

Applications

- A horizontal line on a distance–time graph means the body is at rest.
- A straight line through the origin means the two quantities are directly proportional.
- A curved velocity–time graph means the acceleration is not uniform.

✓ **Points to Remember (Exam Focus)**

- ✓ Distance–time graph: slope = speed.
- ✓ Velocity–time graph: slope = acceleration, area = displacement.
- ✓ A straight line through the origin indicates direct proportionality ($y \propto x$).
- ✓ A steeper slope means a greater rate of change.
- ✓ Negative slope means the quantity is decreasing.

Common Student Mistakes

-  Reading the area under a distance–time graph as if it meant something; only slope is meaningful there.
-  Forgetting that area below the time axis on a velocity–time graph is negative displacement.
-  Taking the slope as y/x instead of $\Delta y/\Delta x$ when the line does not pass through the origin.

Memory Hook

'Slope tells the Speed of change, Area tells the Amount accumulated.'

Advanced Insights

Slope and area are the graphical faces of differentiation and integration. Recognising them at this level makes calculus in higher classes feel like formal notation for something already understood rather than a new subject.

II Quick Recap — Topics Covered So Far

- **Graphs and Their Physical Meaning** — The slope of a graph gives the rate of change of the quantity plotted on the y-axis, and the area under a graph gives the product of the two plotted quantities.

Chapter 0 — End of Chapter Revision

All Memory Hooks

- Algebra and Quadratic Equations in Physics: 'Symbols first, numbers last' — and 'if the units do not match, the equation is wrong'.
- Trigonometry and Vectors for Physics: 'SOH-CAH-TOA' — and 'the Cosine hugs the angle, the Sine stands opposite'.
- Graphs and Their Physical Meaning: 'Slope tells the Speed of change, Area tells the Amount accumulated.'

All Common Mistakes

-  Substituting numbers too early and then being unable to check units.
-  Keeping both roots of a quadratic when one is physically impossible.
-  Cancelling terms across a plus sign instead of across a multiplication.
-  Interchanging sin and cos when resolving — always identify which side the angle is measured from.
-  Adding vector magnitudes arithmetically instead of resolving into components first.
-  Forgetting that the resultant of two equal and opposite vectors is zero, not double.
-  Reading the area under a distance–time graph as if it meant something; only slope is meaningful there.
-  Forgetting that area below the time axis on a velocity–time graph is negative displacement.
-  Taking the slope as y/x instead of $\Delta y/\Delta x$ when the line does not pass through the origin.

Formula Sheet

Formula	Meaning / Use
Quadratic Formula: $x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$	Whenever an equation of motion or energy leads to a squared unknown, e.g. finding time from $s = ut + \frac{1}{2}at^2$.
Resolution of a Vector: $A_x = A \cos \theta$, $A_y = A \sin \theta$	Whenever a force, velocity or displacement acts at an angle and must be split along two perpendicular directions.
Resultant of Perpendicular Vectors: $R = \sqrt{R_x^2 + R_y^2}$	To recombine components into a single vector after adding them axis by axis.

Quick Revision Section

- Algebra and Quadratic Equations in Physics: Physics constantly requires rearranging formulae, solving quadratic equations and handling powers of ten, so fluency in these algebraic tools is the first requirement.
- Trigonometry and Vectors for Physics: Trigonometric ratios resolve a quantity into components along two perpendicular directions, which is the standard way of handling forces and velocities at an angle.
- Graphs and Their Physical Meaning: The slope of a graph gives the rate of change of the quantity plotted on the y-axis, and the area under a graph gives the product of the two plotted quantities.

Chapter 1: Motion

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea of distance, time and speed from earlier classes.
- Ability to plot and read a simple graph.
- Familiarity with SI units — metre, second, kilogram.
- Basic algebra: rearranging an equation to make a chosen symbol the subject.

1.1 Rest, Motion, Distance and Displacement

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Motion is a change of position with respect to a reference point; distance is the total path length covered while displacement is the shortest straight-line change in position.

Real-Life Example

Running one full lap of a 400 m track means a distance of 400 m but a displacement of zero — you finish exactly where you started.

Detailed Explanation

Definition

A body is said to be in motion if its position changes with time with respect to a reference point. Distance is the total length of the path travelled, a scalar. Displacement is the shortest distance from the initial to the final position, a vector.

Key Idea

All motion is relative — a passenger sitting in a moving train is at rest relative to the train but in motion relative to the platform. There is no such thing as absolute rest.

Working Principle

Distance can never be zero or negative for a moving body, but displacement can be zero (a closed path) or negative (motion in the negative direction). The magnitude of displacement is always less than or equal to the distance travelled.

Applications

- GPS navigation reports both route distance and straight-line displacement.
- In a circular track, after one complete round distance = circumference but displacement = 0.
- Displacement equals distance only for motion in a straight line without reversing.

✓ Points to Remember (Exam Focus)

- ✓ Distance is a scalar; displacement is a vector.
- ✓ $|\text{Displacement}| \leq \text{Distance}$, always.
- ✓ Displacement can be zero, positive or negative; distance is always positive.
- ✓ Motion and rest are relative to a chosen reference point.
- ✓ SI unit of both distance and displacement is the metre (m).

⚠ Common Student Mistakes

- ⚠ Treating distance and displacement as interchangeable in a numerical problem.
- ⚠ Saying displacement is 'distance in a straight line' without specifying direction — direction is essential.
- ⚠ Forgetting that a body returning to its start has zero displacement but non-zero distance.

**Memory Hook**

'Distance is the path you Traced; Displacement is the Direct route' — D for Direct, D for Displacement.

**Advanced Insights**

Because displacement is a vector it obeys vector addition, so two successive displacements of 3 m and 4 m can give a resultant anywhere from 1 m to 7 m. Distances, being scalars, would simply add to 7 m in every case.

1.2 Speed, Velocity and Acceleration

🕒 **Estimated time:** 16 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Speed is the rate of covering distance, velocity is the rate of change of displacement, and acceleration is the rate of change of velocity.

**Real-Life Example**

A car's speedometer shows speed only; to know its velocity you also need the direction, and turning a corner at constant speed still counts as acceleration.

**Detailed Explanation****Definition**

Speed = distance/time (scalar). Velocity = displacement/time (vector). Acceleration = change in velocity/time taken (vector). Uniform velocity means equal displacements in equal intervals of time in the same direction.

Key Idea

Velocity changes if either the magnitude or the direction changes. A body moving in a circle at constant speed is continuously accelerating because its direction is constantly changing.

Working Principle

Average speed is total distance divided by total time, which is not generally the same as the average of the individual speeds. Retardation, or deceleration, is simply negative acceleration.

Applications

- Free fall near the earth has a constant acceleration of about 9.8 m/s^2 .
- A body slowing down has acceleration opposite in direction to its velocity.
- Uniform circular motion has constant speed but continuously changing velocity.

✓ Points to Remember (Exam Focus)

- ✓ SI unit of speed and velocity is m/s ; of acceleration, m/s^2 .
- ✓ Average speed = total distance / total time — never the average of two speeds.
- ✓ Acceleration is zero for uniform velocity, positive for speeding up, negative for slowing down.
- ✓ Uniform circular motion: constant speed, changing velocity, non-zero acceleration.
- ✓ $1 \text{ km/h} = 5/18 \text{ m/s}$.

⚠ Common Student Mistakes

- ⚠ Averaging two speeds directly instead of using total distance over total time.
- ⚠ Saying acceleration is zero when a body moves in a circle at constant speed.
- ⚠ Forgetting to convert km/h to m/s before substituting into an equation of motion.

🧠 Memory Hook

'Speed has no direction, Velocity has a Vector' — and 'to convert km/h to m/s , multiply by $5/18$ '.

🚀 Advanced Insights

Instantaneous velocity is the limit of average velocity as the time interval approaches zero — formally the derivative of displacement with respect to time. A speedometer reading is an instantaneous speed, which is why it changes moment to moment.

Formula Section

Average Speed

Difficulty	● Easy ★☆☆
Formula	$v_{\text{avg}} = \text{Total distance} / \text{Total time}$
Meaning of Variables	v_{avg} = average speed in m/s, distance in m, time in s.
When to Use	For any journey in stages, especially where the speed differs in different stages.
Common Mistakes	Taking the arithmetic mean of the two speeds, which is correct only if the two times are equal.
Shortcut / Memory Trick	Equal distances at speeds v_1 and v_2 give average speed $2v_1v_2/(v_1 + v_2)$, the harmonic mean.

Acceleration

Difficulty	● Easy ★☆☆
Formula	$a = (v - u) / t$
Meaning of Variables	a = acceleration in m/s^2 , v = final velocity, u = initial velocity, t = time taken.
When to Use	Whenever a velocity changes over a known interval of time.
Common Mistakes	Reversing u and v , which gives the wrong sign for the acceleration.
Shortcut / Memory Trick	If the answer is negative, the body is slowing down — that is retardation.

II Quick Recap — Topics Covered So Far

- Rest, Motion, Distance and Displacement — Motion is a change of position with respect to a reference point; distance is the total path length covered while displacement is the shortest straight-line change in position.
- Speed, Velocity and Acceleration — Speed is the rate of covering distance, velocity is the rate of change of displacement, and acceleration is the rate of change of velocity.

1.3 Equations of Uniformly Accelerated Motion

 **Estimated time:** 18 mins **Difficulty:** ● Medium ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Three equations connect initial velocity, final velocity, acceleration, time and displacement for motion with constant acceleration.

Real-Life Example

Calculating the length of runway an aircraft needs to reach take-off speed is exactly a $v^2 = u^2 + 2as$ problem.

Detailed Explanation

Definition

The equations of motion are $v = u + at$, $s = ut + \frac{1}{2}at^2$, and $v^2 = u^2 + 2as$, valid only when the acceleration is uniform.

Key Idea

Each equation omits one variable. The first has no s , the second has no v , and the third has no t . Choosing the right equation means identifying which quantity is neither given nor asked for.

Working Principle

A consistent sign convention is essential: choose one direction as positive and give every vector quantity a sign accordingly. For a body thrown upward, taking upward as positive makes g negative.

Applications

- Free fall: substitute $a = g = 9.8 \text{ m/s}^2$ and the same three equations apply.
- Distance travelled in the n th second: $s_n = u + a(2n - 1)/2$.
- Braking distance of a vehicle uses $v^2 = u^2 + 2as$ with $v = 0$.

✓ Points to Remember (Exam Focus)

- ✓ The three equations apply ONLY when acceleration is uniform.
- ✓ $v = u + at$ (no s); $s = ut + \frac{1}{2}at^2$ (no v); $v^2 = u^2 + 2as$ (no t).
- ✓ For a freely falling body, $u = 0$ and $a = g = 9.8 \text{ m/s}^2$.
- ✓ For a body thrown up, velocity at the highest point is zero but acceleration is still g .
- ✓ Use a consistent sign convention for direction throughout a problem.

⚠ Common Student Mistakes

- ⚠ Applying these equations to non-uniform acceleration, where they are invalid.
- ⚠ Taking acceleration as zero at the top of a vertical throw — the velocity is zero, not the acceleration.
- ⚠ Mixing units, such as u in km/h with a in m/s^2 .



Memory Hook

'First has no s , second has no v , third has no t ' — pick the equation that omits what you neither know nor need.



Advanced Insights

The three equations are not independent — the third is obtained by eliminating t between the first two. All three are consequences of a single statement: acceleration is constant, so the velocity–time graph is a straight line.

Formula Section

First Equation of Motion

Difficulty	● Easy ★☆☆
Formula	$v = u + at$
Meaning of Variables	v = final velocity, u = initial velocity, a = acceleration, t = time.
When to Use	When displacement is neither given nor required.
Common Mistakes	Using it when the acceleration is not uniform.
Shortcut / Memory Trick	This is just the definition of acceleration rearranged.

Second Equation of Motion

Difficulty	● Medium ★★☆☆
Formula	$s = ut + \frac{1}{2}at^2$
Meaning of Variables	s = displacement, u = initial velocity, a = acceleration, t = time.
When to Use	When final velocity is neither given nor required.
Common Mistakes	Forgetting the factor of $\frac{1}{2}$, or squaring only the a instead of the t .
Shortcut / Memory Trick	For a body starting from rest, $u = 0$ and it reduces to $s = \frac{1}{2}at^2$.

Third Equation of Motion

Difficulty	● Medium ★★☆☆
Formula	$v^2 = u^2 + 2as$
Meaning of Variables	v = final velocity, u = initial velocity, a = acceleration, s = displacement.
When to Use	When time is neither given nor required — typically braking and stopping-distance problems.
Common Mistakes	Writing $v^2 = u^2 + 2at$, substituting time in place of displacement.
Shortcut / Memory Trick	Set $v = 0$ to find stopping distance directly: $s = -u^2/2a$.

Derivation Section

Derivation: Equations of Motion from the Velocity–Time Graph

Why This Derivation Matters

It shows that all three equations follow from a single straight-line velocity–time graph, so they need not be memorised as unrelated results.

Assumptions Used

- The acceleration is uniform throughout the motion.
- Motion is along a straight line.

Step-by-Step Derivation

Step 1: Draw a velocity–time graph: a straight line from initial velocity u at time 0 to final velocity v at time t .

Step 2: The slope of this line is the acceleration, so $a = (v - u)/t$, which rearranges to $v = u + at$ — the first equation.

Step 3: The area under the graph gives the displacement. The shape is a trapezium of parallel sides u and v and width t , so $s = \frac{1}{2}(u + v)t$.

Step 4: Substituting $v = u + at$ into $s = \frac{1}{2}(u + v)t$ gives $s = \frac{1}{2}(u + u + at)t = ut + \frac{1}{2}at^2$ — the second equation.

Step 5: Eliminating t between $v = u + at$ and $s = \frac{1}{2}(u + v)t$ gives $s = (v^2 - u^2)/2a$, which rearranges to $v^2 = u^2 + 2as$ — the third equation.

Final Result

$$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$$

Exam Perspective

Board papers frequently ask for the graphical derivation of any one of the three — always draw and label the graph first, and state clearly that slope gives acceleration and area gives displacement.

1.4 Uniform Circular Motion

 **Estimated time:** 12 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**    

Quick Summary

A body moving along a circular path at constant speed is in uniform circular motion, which is accelerated motion because the direction of velocity changes continuously.

Real-Life Example

The moon orbiting the earth, a stone whirled on a string, and a cyclist on a circular track are all accelerating even though their speed never changes.

Detailed Explanation

Definition

Uniform circular motion is motion along a circular path with constant speed. The velocity at any instant is directed along the tangent to the circle at that point.

Key Idea

Speed is constant but velocity is not, because direction changes at every instant. A changing velocity means acceleration, and that acceleration points towards the centre — it is called centripetal acceleration.

Working Principle

In one complete revolution the body covers a distance equal to the circumference $2\pi r$ in a time T called the time period, so the speed is $v = 2\pi r/T$. If the string is cut, the body flies off along the tangent, not radially outward.

Applications

- A satellite in a circular orbit around the earth.
- Mud flying off a spinning bicycle wheel leaves tangentially.
- A cyclist leans inward on a curve to obtain the necessary centripetal force.

✓ Points to Remember (Exam Focus)

- ✓ Uniform circular motion has constant speed but changing velocity.
- ✓ It is accelerated motion; the acceleration is directed towards the centre.
- ✓ Velocity at any point is along the tangent to the circle.
- ✓ $v = 2\pi r/T$, where T is the time for one complete revolution.
- ✓ If the string breaks, the body moves along the tangent at that instant.

⚠ Common Student Mistakes

- ⚠ Saying the body flies radially outward when released; it moves along the tangent.
- ⚠ Claiming acceleration is zero because the speed is constant.
- ⚠ Confusing the time period T (per revolution) with the total time of motion.

🧠 Memory Hook

'Constant Speed, Changing Velocity, Centre-seeking acceleration' — and 'cut the string, fly the tangent'.

🚀 Advanced Insights

The centripetal acceleration has magnitude v^2/r . This single expression explains why tight turns at high speed are dangerous — halving the radius or doubling the speed both demand far more force from the tyres than intuition suggests.

Formula Section**📐 Speed in Uniform Circular Motion****Difficulty**

● Easy ★☆☆

Formula	$v = 2\pi r / T$
Meaning of Variables	v = speed, r = radius of the circular path, T = time period of one revolution.
When to Use	To find the speed of any body moving in a circle when the radius and time period are known.
Common Mistakes	Using the diameter in place of the radius, or forgetting the factor 2π .
Shortcut / Memory Trick	$2\pi r$ is the distance for one lap; divide by the time for one lap.

II Quick Recap — Topics Covered So Far

- Equations of Uniformly Accelerated Motion — Three equations connect initial velocity, final velocity, acceleration, time and displacement for motion with constant acceleration.
- Uniform Circular Motion — A body moving along a circular path at constant speed is in uniform circular motion, which is accelerated motion because the direction of velocity changes continuously.

Chapter 1 — End of Chapter Revision

All Memory Hooks

- Rest, Motion, Distance and Displacement: 'Distance is the path you Traced; Displacement is the Direct route' — D for Direct, D for Displacement.
- Speed, Velocity and Acceleration: 'Speed has no direction, Velocity has a Vector' — and 'to convert km/h to m/s, multiply by 5/18'.
- Equations of Uniformly Accelerated Motion: 'First has no s, second has no v, third has no t' — pick the equation that omits what you neither know nor need.
- Uniform Circular Motion: 'Constant Speed, Changing Velocity, Centre-seeking acceleration' — and 'cut the string, fly the tangent'.

All Common Mistakes

- ⚠ Treating distance and displacement as interchangeable in a numerical problem.
- ⚠ Saying displacement is 'distance in a straight line' without specifying direction — direction is essential.
- ⚠ Forgetting that a body returning to its start has zero displacement but non-zero distance.
- ⚠ Averaging two speeds directly instead of using total distance over total time.
- ⚠ Saying acceleration is zero when a body moves in a circle at constant speed.
- ⚠ Forgetting to convert km/h to m/s before substituting into an equation of motion.
- ⚠ Applying these equations to non-uniform acceleration, where they are invalid.
- ⚠ Taking acceleration as zero at the top of a vertical throw — the velocity is zero, not the acceleration.
- ⚠ Mixing units, such as u in km/h with a in m/s².
- ⚠ Saying the body flies radially outward when released; it moves along the tangent.
- ⚠ Claiming acceleration is zero because the speed is constant.
- ⚠ Confusing the time period T (per revolution) with the total time of motion.

Formula Sheet

Formula	Meaning / Use
Average Speed: $v_{\text{avg}} = \text{Total distance} / \text{Total time}$	For any journey in stages, especially where the speed differs in different stages.
Acceleration: $a = (v - u) / t$	Whenever a velocity changes over a known interval of time.
First Equation of Motion: $v = u + at$	When displacement is neither given nor required.
Second Equation of Motion: $s = ut + \frac{1}{2}at^2$	When final velocity is neither given nor required.

Third Equation of Motion: $v^2 = u^2 + 2as$	When time is neither given nor required — typically braking and stopping-distance problems.
Speed in Uniform Circular Motion: $v = 2\pi r / T$	To find the speed of any body moving in a circle when the radius and time period are known.

Quick Revision Section

- Rest, Motion, Distance and Displacement: Motion is a change of position with respect to a reference point; distance is the total path length covered while displacement is the shortest straight-line change in position.
- Speed, Velocity and Acceleration: Speed is the rate of covering distance, velocity is the rate of change of displacement, and acceleration is the rate of change of velocity.
- Equations of Uniformly Accelerated Motion: Three equations connect initial velocity, final velocity, acceleration, time and displacement for motion with constant acceleration.
- Uniform Circular Motion: A body moving along a circular path at constant speed is in uniform circular motion, which is accelerated motion because the direction of velocity changes continuously.

Chapter 2: Force and Laws of Motion

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Equations of uniformly accelerated motion from the previous chapter.
- Idea of mass as the quantity of matter in a body.
- Understanding of velocity and acceleration as vectors.
- Familiarity with the units newton, kilogram and second.

2.1 Force, Balanced and Unbalanced Forces

 **Estimated time:** 13 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

A force is a push or pull that can change the state of rest or motion, the direction of motion, or the shape of a body; only an unbalanced force produces acceleration.

Real-Life Example

In a tug of war neither team moves while the pulls are balanced — the rope only shifts the instant one side pulls harder.

Detailed Explanation

Definition

Force is a push or a pull acting on a body. Balanced forces are forces whose resultant is zero; unbalanced forces have a non-zero resultant.

Key Idea

Balanced forces cannot change the state of motion — they can only change the shape of a body. Every change in speed or direction requires an unbalanced force.

Working Principle

Forces are vectors, so they combine by vector addition. Forces in the same direction add; forces in opposite directions subtract, and the resultant acts in the direction of the larger force.

Applications

- A book resting on a table has weight balanced exactly by the normal reaction.
- Squeezing a rubber ball with two equal opposite forces changes shape without producing motion.
- A moving vehicle needs continuous engine force only because friction opposes it.

✓ **Points to Remember (Exam Focus)**

- ✓ SI unit of force is the newton (N); $1 \text{ N} = 1 \text{ kg}\cdot\text{m}/\text{s}^2$.
- ✓ Balanced forces $\rightarrow$ no change in state of motion; may change shape.
- ✓ Unbalanced forces $\rightarrow$ acceleration in the direction of the resultant.
- ✓ Force is a vector quantity — magnitude and direction both matter.
- ✓ Contact forces: friction, normal reaction, tension. Non-contact: gravitational, magnetic, electrostatic.

⚠ Common Student Mistakes

- ⚠ Assuming a moving body must have a net force acting on it — it may be moving at constant velocity with zero net force.
- ⚠ Adding forces arithmetically without regard to direction.
- ⚠ Thinking balanced forces have no effect at all; they can deform a body.

🧠 Memory Hook

'Balanced = no change in motion; Unbalanced = acceleration' — the resultant is what matters, never the individual forces.

🚀 Advanced Insights

The distinction between contact and non-contact forces dissolves at the microscopic level: normal reaction and friction are ultimately electromagnetic repulsion between the electron clouds of atoms that never truly touch.

2.2 Newton's First Law and Inertia

🕒 **Estimated time:** 14 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

🗣 Quick Summary

A body continues in its state of rest or uniform motion in a straight line unless acted on by an unbalanced external force — the property responsible is called inertia, and mass is its measure.

🌐 Real-Life Example

When a bus brakes suddenly, passengers lurch forward because their bodies continue moving even after the bus has stopped.

📖 Detailed Explanation

Definition

Newton's first law of motion states that a body at rest remains at rest, and a body in uniform motion continues in uniform motion in a straight line, unless compelled by an unbalanced external force to change that state. Inertia is the tendency of a body to resist a change in its state of motion.

Key Idea

The first law is also called the law of inertia, and it defines force as that which changes motion. Mass is the quantitative measure of inertia — the greater the mass, the harder it is to start, stop or turn a body.

Working Principle

There are three kinds of inertia: inertia of rest (a body stays at rest), inertia of motion (a body keeps moving), and inertia of direction (a body keeps moving in a straight line).

Applications

- Dust flies off a carpet when beaten — the carpet moves, the dust stays put (inertia of rest).
- An athlete runs before a long jump to make use of inertia of motion.
- A passenger falls outward when a vehicle turns sharply (inertia of direction).

✓ Points to Remember (Exam Focus)

- ✓ First law defines force and introduces inertia; it is also called the law of inertia.
- ✓ Mass is the measure of inertia — greater mass means greater inertia.
- ✓ Inertia of rest, of motion and of direction are the three types.
- ✓ No force is needed to keep a body moving at constant velocity — only to change it.
- ✓ Galileo's experiments with inclined planes led to this law.

⚠ Common Student Mistakes

- ⚠ Saying inertia is a force — inertia is a property, not a force.
- ⚠ Believing a heavier body always falls faster; mass affects inertia, not free-fall acceleration.
- ⚠ Thinking a constantly moving body must have a force pushing it forward.



Memory Hook

'Lazy law' — objects are lazy and refuse to change what they are already doing; mass measures how lazy.



Advanced Insights

The first law holds only in an inertial frame — one that is not itself accelerating. In an accelerating bus the law appears violated, and physicists restore it by introducing pseudo-forces such as the centrifugal force.

II Quick Recap — Topics Covered So Far

- **Force, Balanced and Unbalanced Forces** — A force is a push or pull that can change the state of rest or motion, the direction of motion, or the shape of a body; only an unbalanced force produces acceleration.
- **Newton's First Law and Inertia** — A body continues in its state of rest or uniform motion in a straight line unless acted on by an unbalanced external force — the property responsible is called inertia, and mass is its measure.

2.3 Newton's Second Law and Momentum

Estimated time: 17 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force, which gives the familiar $F = ma$.

Real-Life Example

A cricketer moves their hands backward while catching a ball — increasing the time of impact reduces the force on the hands for the same change in momentum.

Detailed Explanation

Definition

Momentum is the product of mass and velocity, $p = mv$, a vector with SI unit $\text{kg}\cdot\text{m/s}$. Newton's second law states that the rate of change of momentum is proportional to the applied force and occurs in the direction of the force.

Key Idea

$F = ma$ is only the special case of the second law for constant mass. The general statement $F = \Delta p / \Delta t$ explains why the same change in momentum can be produced by a small force over a long time or a large force over a short time.

Working Principle

Impulse is the product of force and the time for which it acts, and it equals the change in momentum: $F \times t = mv - mu$. This is the principle behind every safety device that lengthens the time of an impact.

Applications

- Airbags and crumple zones increase collision time, reducing the force on passengers.
- A high jumper lands on a thick foam mattress to increase the stopping time.
- Athletes bend their knees on landing for exactly the same reason.

Points to Remember (Exam Focus)

- ✓ Momentum $p = mv$, a vector; SI unit $\text{kg}\cdot\text{m/s}$.

- ✓ $F = \Delta p / \Delta t$ is the general form; $F = ma$ applies when mass is constant.
- ✓ 1 newton is the force that gives a 1 kg mass an acceleration of 1 m/s^2 .
- ✓ Impulse = $F \times t$ = change in momentum.
- ✓ For the same change in momentum, a longer contact time means a smaller force.

⚠ Common Student Mistakes

- ⚠ Using $F = ma$ when the mass is changing, such as for a rocket.
- ⚠ Forgetting that momentum is a vector, so a reversal of direction doubles the change.
- ⚠ Confusing momentum (mv) with kinetic energy ($\frac{1}{2}mv^2$).

🧠 Memory Hook

'Force = rate of change of momentum' — and 'more time, less force' for every safety device.

🚀 Advanced Insights

Momentum, not velocity, is the quantity nature conserves. Newton originally wrote his second law in terms of momentum, and that form survives unchanged into relativity, where $F = ma$ does not.

Formula Section

📐 Momentum

Difficulty	● Easy ★☆☆
Formula	$p = m \times v$
Meaning of Variables	p = momentum in $\text{kg}\cdot\text{m/s}$, m = mass in kg , v = velocity in m/s .
When to Use	Whenever collisions, impulse or conservation problems are involved.
Common Mistakes	Ignoring the direction, which matters because momentum is a vector.
Shortcut / Memory Trick	A light fast body and a heavy slow body can have the same momentum.

📐 Newton's Second Law

Difficulty	● Medium ★★★
Formula	$F = ma = (mv - mu) / t$
Meaning of Variables	F = net force in N , m = mass in kg , a = acceleration in m/s^2 , u and v = initial and final velocities.

When to Use	To find force, mass or acceleration when the other two are known, or to find force from a momentum change.
Common Mistakes	Substituting a single force instead of the NET force acting on the body.
Shortcut / Memory Trick	Always draw the free-body diagram and add up forces before using $F = ma$.

Impulse

Difficulty	● Medium ★★☆☆
Formula	$J = F \times t = \Delta p$
Meaning of Variables	J = impulse in N·s, F = force, t = time of action, Δp = change in momentum.
When to Use	For short-duration collisions where the force is large and the contact time small.
Common Mistakes	Forgetting that impulse and momentum have the same units and are numerically equal.
Shortcut / Memory Trick	Impulse is the area under a force–time graph.

Derivation Section

Derivation: $F = ma$ from Newton's Second Law

Why This Derivation Matters

It shows that the familiar formula is not a separate law but a direct consequence of the momentum statement, valid only when the mass is constant.

Assumptions Used

- The mass of the body remains constant during the motion.
- The force acts along the direction of motion.

Step-by-Step Derivation

Step 1: Let a body of mass m have initial velocity u ; after time t under a constant force its velocity is v .

Step 2: Initial momentum = mu , final momentum = mv , so the change in momentum is $mv - mu = m(v - u)$.

Step 3: The rate of change of momentum is therefore $m(v - u)/t$.

Step 4: By the second law, force is proportional to this rate: $F \propto m(v - u)/t$.

Step 5: Since $(v - u)/t$ is the acceleration a , $F \propto ma$, and choosing the newton so that the constant of proportionality is 1 gives $F = ma$.

Final Result

$$F = ma$$

Exam Perspective

State the second law in words first, then show the momentum change step by step. Examiners look specifically for the sentence that defines the newton so that $k = 1$.

2.4 Newton's Third Law and Conservation of Momentum

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

To every action there is an equal and opposite reaction, and it follows that the total momentum of an isolated system remains constant.

Real-Life Example

A rifle recoils backward when fired, and a rocket rises because it pushes exhaust gases downward — both are the third law and momentum conservation in action.

Detailed Explanation

Definition

Newton's third law states that to every action there is an equal and opposite reaction, and the two act on two different bodies. The law of conservation of momentum states that in the absence of an external force, the total momentum of a system remains constant.

Key Idea

Action and reaction are equal in magnitude and opposite in direction but never cancel, because they act on different bodies. This is the single most misunderstood point in the whole chapter.

Working Principle

For two bodies colliding, $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$. The forces they exert on each other are equal and opposite and act for the same time, so the momentum gained by one equals the momentum lost by the other.

Applications

- Recoil of a gun: the gun's backward momentum equals the bullet's forward momentum.
- Rocket propulsion: exhaust gases pushed down, rocket pushed up.
- Walking: we push the ground backward, the ground pushes us forward.

✓ Points to Remember (Exam Focus)

- ✓ Action and reaction act on DIFFERENT bodies, so they never cancel out.
- ✓ They are equal in magnitude and opposite in direction, and act simultaneously.
- ✓ Total momentum before collision = total momentum after collision, if no external force acts.
- ✓ Recoil velocity of a gun = (mass of bullet × velocity of bullet) / mass of gun.
- ✓ Momentum is conserved in all collisions; kinetic energy is conserved only in elastic ones.

Common Student Mistakes

- ⚠️ Saying action and reaction cancel each other — they act on different bodies.
- ⚠️ Forgetting to give velocities opposite signs when bodies move in opposite directions.
- ⚠️ Assuming kinetic energy is always conserved in a collision; it is not, except in elastic collisions.



Memory Hook

'Equal, Opposite, but on Different bodies' — and 'gun goes back because bullet goes forward'.



Advanced Insights

Conservation of momentum is more fundamental than Newton's laws themselves — it follows from the fact that the laws of physics are the same everywhere in space. This deep link between symmetry and conservation is Noether's theorem.

Formula Section



Conservation of Momentum

Difficulty	● Medium ★★☆☆
Formula	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
Meaning of Variables	m_1, m_2 = masses; u_1, u_2 = velocities before collision; v_1, v_2 = velocities after collision.
When to Use	For any collision, explosion or recoil where no external force acts on the system.
Common Mistakes	Using the same sign for velocities in opposite directions.
Shortcut / Memory Trick	For a body initially at rest that explodes, total momentum stays zero, so the fragments move in exactly opposite directions.

Derivation Section



Derivation: Law of Conservation of Momentum from the Third Law

Why This Derivation Matters

It proves that momentum conservation is not an independent assumption but a direct consequence of Newton's third law.

Assumptions Used

- The two bodies form an isolated system with no external force acting.
- The collision lasts for a time t , the same for both bodies.

Step-by-Step Derivation

Step 1: Two bodies of masses m_1 and m_2 move with initial velocities u_1 and u_2 along the same line, with $u_1 > u_2$ so that they collide.

Step 2: During the collision body 1 exerts a force F_{21} on body 2, and body 2 exerts F_{12} on body 1. By the third law, $F_{12} = -F_{21}$.

Step 3: The force on each equals its rate of change of momentum: $F_{12} = m_1(v_1 - u_1)/t$ and $F_{21} = m_2(v_2 - u_2)/t$.

Step 4: Substituting into $F_{12} = -F_{21}$ gives $m_1(v_1 - u_1)/t = -m_2(v_2 - u_2)/t$.

Step 5: Cancelling t and rearranging gives $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ — the total momentum before equals the total momentum after.

Final Result

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Exam Perspective

This is a very frequently asked derivation. Marks are given specifically for stating the third law relation $F_{12} = -F_{21}$ and for noting that the time of contact is the same for both bodies.

II Quick Recap — Topics Covered So Far

- Newton's Second Law and Momentum — The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force, which gives the familiar $F = ma$.
- Newton's Third Law and Conservation of Momentum — To every action there is an equal and opposite reaction, and it follows that the total momentum of an isolated system remains constant.

Chapter 2 — End of Chapter Revision

All Memory Hooks

- Force, Balanced and Unbalanced Forces: 'Balanced = no change in motion; Unbalanced = acceleration' — the resultant is what matters, never the individual forces.
- Newton's First Law and Inertia: 'Lazy law' — objects are lazy and refuse to change what they are already doing; mass measures how lazy.
- Newton's Second Law and Momentum: 'Force = rate of change of momentum' — and 'more time, less force' for every safety device.
- Newton's Third Law and Conservation of Momentum: 'Equal, Opposite, but on Different bodies' — and 'gun goes back because bullet goes forward'.

All Common Mistakes

-  Assuming a moving body must have a net force acting on it — it may be moving at constant velocity with zero net force.
-  Adding forces arithmetically without regard to direction.
-  Thinking balanced forces have no effect at all; they can deform a body.
-  Saying inertia is a force — inertia is a property, not a force.
-  Believing a heavier body always falls faster; mass affects inertia, not free-fall acceleration.
-  Thinking a constantly moving body must have a force pushing it forward.
-  Using $F = ma$ when the mass is changing, such as for a rocket.
-  Forgetting that momentum is a vector, so a reversal of direction doubles the change.
-  Confusing momentum (mv) with kinetic energy ($\frac{1}{2}mv^2$).
-  Saying action and reaction cancel each other — they act on different bodies.
-  Forgetting to give velocities opposite signs when bodies move in opposite directions.
-  Assuming kinetic energy is always conserved in a collision; it is not, except in elastic collisions.

Formula Sheet

Formula	Meaning / Use
Momentum: $p = m \times v$	Whenever collisions, impulse or conservation problems are involved.
Newton's Second Law: $F = ma = (mv - mu) / t$	To find force, mass or acceleration when the other two are known, or to find force from a momentum change.

Impulse: $J = F \times t = \Delta p$	For short-duration collisions where the force is large and the contact time small.
Conservation of Momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	For any collision, explosion or recoil where no external force acts on the system.

Quick Revision Section

- **Force, Balanced and Unbalanced Forces:** A force is a push or pull that can change the state of rest or motion, the direction of motion, or the shape of a body; only an unbalanced force produces acceleration.
- **Newton's First Law and Inertia:** A body continues in its state of rest or uniform motion in a straight line unless acted on by an unbalanced external force — the property responsible is called inertia, and mass is its measure.
- **Newton's Second Law and Momentum:** The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force, which gives the familiar $F = ma$.
- **Newton's Third Law and Conservation of Momentum:** To every action there is an equal and opposite reaction, and it follows that the total momentum of an isolated system remains constant.

Chapter 3: Gravitation

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Newton's three laws of motion and the idea of force.
- Equations of uniformly accelerated motion.
- Difference between mass and weight in everyday language.
- Idea that objects fall towards the earth.

3.1 Universal Law of Gravitation

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Every body in the universe attracts every other body with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Real-Life Example

The same force that makes an apple fall keeps the moon in orbit — Newton's insight was that these are not two phenomena but one.

Detailed Explanation

Definition

The universal law of gravitation states that the force of attraction between two bodies is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

Key Idea

The inverse-square dependence is what makes gravity weak at large distances: doubling the separation reduces the force to one quarter, not one half. G , the universal gravitational constant, is the same everywhere in the universe.

Working Principle

The two forces on the two bodies form an action–reaction pair — the earth pulls an apple with exactly the force the apple pulls the earth. The earth does not visibly move only because its mass is enormous.

Applications

- Explains the motion of planets around the sun and the moon around the earth.
- Explains the occurrence of tides due to the moon and the sun.
- Explains why the value of g varies slightly from place to place on earth.

✓ Points to Remember (Exam Focus)

- ✓ $F = Gm_1m_2/r^2$, where $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$.
- ✓ G is a universal constant — the same everywhere in the universe.
- ✓ Gravitational force is always attractive and acts along the line joining the centres.
- ✓ The two forces form an action–reaction pair of equal magnitude.
- ✓ Distance r is measured between the CENTRES of the two bodies.

⚠ Common Student Mistakes

- ⚠ Using the distance between surfaces instead of between centres.
- ⚠ Confusing G (universal constant, 6.67×10^{-11}) with g (acceleration due to gravity, 9.8 m/s^2).
- ⚠ Forgetting to square the distance in the denominator.



Memory Hook

'Big G is Global and constant; small g is Ground-level and varies.'



Advanced Insights

Cavendish measured G in 1798 using a torsion balance, an experiment sensitive enough to detect the pull between lead spheres in a laboratory. Because he could then calculate the earth's mass, the experiment is often called 'weighing the earth'.

Formula Section



Universal Law of Gravitation

Difficulty	● Medium ★★☆☆
Formula	$F = G m_1 m_2 / r^2$
Meaning of Variables	F = gravitational force in N, $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$, m_1 and m_2 = masses in kg, r = distance between centres in m.
When to Use	To find the attraction between any two bodies, including planets, satellites and everyday objects.
Common Mistakes	Forgetting to square r , or using the radius of only one body instead of the centre-to-centre distance.
Shortcut / Memory Trick	Double the distance → force falls to one quarter; triple it → one ninth.

3.2 Acceleration Due to Gravity and Free Fall

🕒 **Estimated time:** 14 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔍 Quick Summary

All bodies falling freely near the earth accelerate at the same rate g , independent of their mass, because a larger mass experiences a proportionally larger force.

🌐 Real-Life Example

On the airless moon, Apollo 15's David Scott dropped a hammer and a feather together and both landed at the same instant.

📖 Detailed Explanation

Definition

Free fall is motion under the influence of gravity alone, with no other force such as air resistance. The acceleration produced is the acceleration due to gravity, g , about 9.8 m/s^2 near the earth's surface.

Key Idea

Equating $F = mg$ with $F = GMm/R^2$ gives $g = GM/R^2$, in which the mass of the falling body cancels out entirely. This is why all bodies fall at the same rate in a vacuum.

Working Principle

The value of g decreases with altitude and with depth, and increases slightly from the equator towards the poles because the earth is not a perfect sphere and it rotates.

Applications

- g on the moon is about $1/6$ of g on the earth, so a body weighs one-sixth as much there.
- The equations of motion apply to free fall with a replaced by g .
- In a satellite everything is in free fall together, producing apparent weightlessness.

✓ Points to Remember (Exam Focus)

- ✓ $g = GM/R^2$, where M and R are the earth's mass and radius.
- ✓ g is independent of the mass of the falling body.
- ✓ $g \approx 9.8 \text{ m/s}^2$ at the earth's surface; it is maximum at the poles and minimum at the equator.
- ✓ g decreases with both altitude and depth, and is zero at the centre of the earth.
- ✓ For a body thrown upward, g acts downward, so $a = -9.8 \text{ m/s}^2$.

⚠ Common Student Mistakes

- ⚠ Believing heavier bodies fall faster — they do so only because of air resistance, not gravity.
- ⚠ Taking g as positive for a body moving upward without adjusting the sign convention.

⚠ Thinking g is zero for an astronaut in orbit; g is nearly full strength, they are simply in free fall.

Memory Hook

'The mass cancels' — that one algebraic fact is the whole reason a feather and a hammer land together.

Advanced Insights

Astronauts float not because gravity has vanished but because they and the station fall around the earth together. True weightlessness would require being infinitely far from all masses, which never happens.

Formula Section

Acceleration Due to Gravity

Difficulty	● Medium ★★☆☆
Formula	$g = GM / R^2$
Meaning of Variables	g = acceleration due to gravity, G = gravitational constant, M = mass of the earth, R = radius of the earth.
When to Use	To calculate g for the earth or for any other planet whose mass and radius are known.
Common Mistakes	Using the distance from the surface instead of the radius from the centre.
Shortcut / Memory Trick	The mass of the falling body never appears — that is the point of the formula.

Equations of Motion Under Gravity

Difficulty	● Medium ★★☆☆
Formula	$v = u + gt$, $h = ut + \frac{1}{2}gt^2$, $v^2 = u^2 + 2gh$
Meaning of Variables	u = initial velocity, v = final velocity, h = height, t = time, $g = 9.8 \text{ m/s}^2$.
When to Use	For any freely falling body or body thrown vertically upward.
Common Mistakes	Using $+g$ for upward motion; take one direction as positive and stay consistent.
Shortcut / Memory Trick	For a body dropped from rest, $u = 0$, so $h = \frac{1}{2}gt^2$.

II Quick Recap — Topics Covered So Far

- Universal Law of Gravitation — Every body in the universe attracts every other body with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
- Acceleration Due to Gravity and Free Fall — All bodies falling freely near the earth accelerate at the same rate g , independent of their mass, because a larger mass experiences a proportionally larger force.

3.3 Mass and Weight

🕒 **Estimated time:** 12 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Mass is the quantity of matter in a body and never changes, while weight is the gravitational force on it and changes from place to place.

Real-Life Example

An astronaut's mass is identical on earth and on the moon, but their weight on the moon is only one-sixth as much.

Detailed Explanation

Definition

Mass is the amount of matter contained in a body, a scalar measured in kilograms. Weight is the force with which the earth attracts a body, a vector measured in newtons, given by $W = mg$.

Key Idea

Mass is a fundamental property of a body; weight is a force that depends on where the body is. Confusing the two is the single most common error in this chapter.

Working Principle

Because weight is mg and g varies with location, the same body weighs slightly more at the poles than at the equator, one-sixth as much on the moon, and nothing at all in free fall.

Applications

- A beam balance compares masses and gives the same reading everywhere.
- A spring balance measures weight and reads differently on the moon.
- $1 \text{ kg wt} = 9.8 \text{ N}$ — the weight of a 1 kg mass on the earth's surface.

✓ **Points to Remember (Exam Focus)**

- ✓ Mass: scalar, kilogram, constant everywhere; Weight: vector, newton, varies with location.
- ✓ $W = mg$, so weight is zero where g is zero.
- ✓ Weight on the moon = $(1/6) \times$ weight on the earth.

- ✓ A beam balance measures mass; a spring balance measures weight.
- ✓ Mass can never be zero for a material body; weight can be.

⚠ Common Student Mistakes

- ⚠ Giving weight in kilograms — the SI unit of weight is the newton.
- ⚠ Saying an astronaut is weightless because they have no mass.
- ⚠ Thinking a beam balance would read differently on the moon; it would not.

🧠 Memory Hook

'Mass is what you are; Weight is what the planet makes of you.'

🚀 Advanced Insights

That gravitational mass (in $W = mg$) and inertial mass (in $F = ma$) are numerically identical is not obvious and has been verified to extraordinary precision. Einstein elevated this coincidence into the equivalence principle, the foundation of general relativity.

Formula Section

📏 Weight

Difficulty	● Easy ★☆☆
Formula	$W = m \times g$
Meaning of Variables	W = weight in newtons, m = mass in kg, g = acceleration due to gravity in m/s^2 .
When to Use	To convert between mass and weight, or to compare weights on different planets.
Common Mistakes	Expressing the answer in kg instead of N.
Shortcut / Memory Trick	Multiply mass by 10 for a quick estimate of weight in newtons on the earth.

3.4 Thrust, Pressure and Archimedes' Principle

🕒 Estimated time: 16 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

🔑 Quick Summary

Pressure is thrust per unit area, fluids exert an upward buoyant force on immersed bodies, and Archimedes' principle states that this upthrust equals the weight of fluid displaced.



Real-Life Example

A sharp knife cuts easily because the same force acts over a tiny area, and a huge steel ship floats because it displaces an enormous weight of water.



Detailed Explanation

Definition

Thrust is the force acting normally on a surface. Pressure is thrust per unit area, $P = F/A$, measured in pascals. Buoyancy is the upward force exerted by a fluid on a body immersed in it.

Key Idea

Archimedes' principle states that a body immersed wholly or partially in a fluid experiences an upthrust equal to the weight of the fluid it displaces. Whether the body floats or sinks depends on how this upthrust compares with its weight.

Working Principle

If the density of the body is less than that of the fluid, it floats; if greater, it sinks; if equal, it stays suspended. Relative density is the ratio of the density of a substance to the density of water and has no unit.

Applications

- Ships are hollow so that their average density is less than that of water.
- A hydrometer measures the relative density of liquids such as milk.
- A body weighs less in water — the apparent loss of weight equals the upthrust.

✓ Points to Remember (Exam Focus)

- ✓ Pressure $P = F/A$; SI unit is the pascal (Pa), $1 \text{ Pa} = 1 \text{ N/m}^2$.
- ✓ Pressure increases with depth in a liquid and acts equally in all directions.
- ✓ Archimedes' principle: upthrust = weight of fluid displaced.
- ✓ Float if $\rho_{\text{body}} < \rho_{\text{fluid}}$; sink if $\rho_{\text{body}} > \rho_{\text{fluid}}$.
- ✓ Relative density = density of substance / density of water; it has no unit.

⚠ Common Student Mistakes

- ⚠ Confusing thrust (a force in newtons) with pressure (force per unit area in pascals).
- ⚠ Thinking upthrust depends on the weight of the body rather than the weight of the fluid displaced.
- ⚠ Giving relative density a unit — it is a pure ratio.

 Memory Hook

'Same force, smaller area, bigger pressure' — and 'Upthrust = weight of the water pushed aside'.

 Advanced Insights

Buoyancy arises because pressure increases with depth, so the upward push on the bottom of a submerged body exceeds the downward push on its top. Archimedes' principle is a consequence of that pressure difference, not a separate law.

Formula Section

 Pressure

Difficulty	● Easy ★☆☆
Formula	$P = F / A$
Meaning of Variables	P = pressure in Pa, F = thrust (normal force) in N, A = area in m^2 .
When to Use	Whenever a force is distributed over a surface — nails, knives, snowshoes, foundations.
Common Mistakes	Using area in cm^2 without converting to m^2 , which gives an answer 10,000 times too large.
Shortcut / Memory Trick	Reduce the area to increase the pressure — that is why blades are sharpened.

 Pressure at Depth in a Liquid

Difficulty	● Medium ★★★
Formula	$P = h \rho g$
Meaning of Variables	P = pressure due to the liquid column, h = depth, ρ = density of liquid, g = acceleration due to gravity.
When to Use	To find the pressure at any depth in a liquid, e.g. at the base of a dam.
Common Mistakes	Including the area of the container — pressure at depth does not depend on it.
Shortcut / Memory Trick	Pressure depends only on depth, density and g — never on the shape of the vessel.

 Archimedes' Principle

Difficulty	● Medium ★★★
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Formula	Upthrust = $V \rho g$ = weight of fluid displaced
Meaning of Variables	V = volume of fluid displaced, ρ = density of the fluid, g = acceleration due to gravity.
When to Use	For all floating, sinking and apparent-weight problems.
Common Mistakes	Using the density of the body instead of the density of the fluid.
Shortcut / Memory Trick	Apparent loss of weight in a fluid is exactly equal to the upthrust.

II Quick Recap — Topics Covered So Far

- **Mass and Weight** — Mass is the quantity of matter in a body and never changes, while weight is the gravitational force on it and changes from place to place.
- **Thrust, Pressure and Archimedes' Principle** — Pressure is thrust per unit area, fluids exert an upward buoyant force on immersed bodies, and Archimedes' principle states that this upthrust equals the weight of fluid displaced.

Chapter 3 — End of Chapter Revision

All Memory Hooks

- Universal Law of Gravitation: 'Big G is Global and constant; small g is Ground-level and varies.'
- Acceleration Due to Gravity and Free Fall: 'The mass cancels' — that one algebraic fact is the whole reason a feather and a hammer land together.
- Mass and Weight: 'Mass is what you are; Weight is what the planet makes of you.'
- Thrust, Pressure and Archimedes' Principle: 'Same force, smaller area, bigger pressure' — and 'Upthrust = weight of the water pushed aside'.

All Common Mistakes

-  Using the distance between surfaces instead of between centres.
-  Confusing G (universal constant, 6.67×10^{-11}) with g (acceleration due to gravity, 9.8 m/s^2).
-  Forgetting to square the distance in the denominator.
-  Believing heavier bodies fall faster — they do so only because of air resistance, not gravity.
-  Taking g as positive for a body moving upward without adjusting the sign convention.
-  Thinking g is zero for an astronaut in orbit; g is nearly full strength, they are simply in free fall.
-  Giving weight in kilograms — the SI unit of weight is the newton.
-  Saying an astronaut is weightless because they have no mass.
-  Thinking a beam balance would read differently on the moon; it would not.
-  Confusing thrust (a force in newtons) with pressure (force per unit area in pascals).
-  Thinking upthrust depends on the weight of the body rather than the weight of the fluid displaced.
-  Giving relative density a unit — it is a pure ratio.

Formula Sheet

Formula	Meaning / Use
Universal Law of Gravitation: $F = G m_1 m_2 / r^2$	To find the attraction between any two bodies, including planets, satellites and everyday objects.
Acceleration Due to Gravity: $g = GM / R^2$	To calculate g for the earth or for any other planet whose mass and radius are known.
Equations of Motion Under Gravity: $v = u + gt$, $h = ut + \frac{1}{2}gt^2$, $v^2 = u^2 + 2gh$	For any freely falling body or body thrown vertically upward.
Weight: $W = m \times g$	To convert between mass and weight, or to compare weights on different planets.

Pressure: $P = F / A$	Whenever a force is distributed over a surface — nails, knives, snowshoes, foundations.
Pressure at Depth in a Liquid: $P = h \rho g$	To find the pressure at any depth in a liquid, e.g. at the base of a dam.
Archimedes' Principle: Upthrust = $V \rho g$ = weight of fluid displaced	For all floating, sinking and apparent-weight problems.

Quick Revision Section

- Universal Law of Gravitation: Every body in the universe attracts every other body with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
- Acceleration Due to Gravity and Free Fall: All bodies falling freely near the earth accelerate at the same rate g , independent of their mass, because a larger mass experiences a proportionally larger force.
- Mass and Weight: Mass is the quantity of matter in a body and never changes, while weight is the gravitational force on it and changes from place to place.
- Thrust, Pressure and Archimedes' Principle: Pressure is thrust per unit area, fluids exert an upward buoyant force on immersed bodies, and Archimedes' principle states that this upthrust equals the weight of fluid displaced.

Chapter 4: Work and Energy

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Newton's laws of motion and the concept of force.
- Equations of uniformly accelerated motion.
- Idea of resolving a force into components.
- Familiarity with the joule as a unit of energy.

4.1 Work Done by a Force

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Work is done only when a force produces a displacement in its own direction, and it is measured as the product of the force and the displacement along the force.

Real-Life Example

Holding a heavy suitcase while standing still is exhausting but does zero work in physics, because there is no displacement.

Detailed Explanation

Definition

Work done by a constant force is the product of the magnitude of the force and the displacement in the direction of the force: $W = F s \cos \theta$, where θ is the angle between force and displacement.

Key Idea

Two conditions must both be met for work to be done: a force must act, and the body must be displaced. If either is missing, or if the displacement is perpendicular to the force, the work done is zero.

Working Principle

Work is positive when the force has a component along the displacement, negative when it opposes the displacement (as friction does), and zero when the force is perpendicular to the displacement.

Applications

- A coolie carrying a load on their head along a level platform does zero work against gravity.
- Friction always does negative work, converting kinetic energy into heat.
- The centripetal force in circular motion does zero work, being always perpendicular to velocity.

Points to Remember (Exam Focus)

- ✓ $W = F s \cos \theta$; SI unit is the joule (J), $1 \text{ J} = 1 \text{ N}\cdot\text{m}$.
- ✓ Work is a scalar quantity even though force and displacement are vectors.
- ✓ Zero work: no displacement, no force, or force perpendicular to displacement.
- ✓ Negative work is done by a force opposing the motion, such as friction.
- ✓ 1 joule is the work done when a force of 1 N moves a body 1 m in its own direction.

⚠ Common Student Mistakes

- ⚠ Saying work is done when a body is merely held stationary — displacement is required.
- ⚠ Omitting $\cos \theta$ when the force acts at an angle to the displacement.
- ⚠ Treating work as a vector; it is a scalar.

🧠 Memory Hook

'No displacement, no work' — and 'perpendicular force does nothing at all'.

🚀 Advanced Insights

Because the centripetal force does no work, a satellite maintains its speed indefinitely without any energy input. This is why orbits are stable and why a planet needs no fuel to keep circling the sun.

Formula Section

📐 Work Done

Difficulty	● Medium ★★☆☆
Formula	$W = F s \cos \theta$
Meaning of Variables	W = work in joules, F = force in N, s = displacement in m, θ = angle between force and displacement.
When to Use	For any constant force producing a displacement, including forces at an angle.
Common Mistakes	Forgetting $\cos \theta$, or using the total distance rather than the displacement along the force.
Shortcut / Memory Trick	$\theta = 0^\circ$ gives $W = Fs$; $\theta = 90^\circ$ gives $W = 0$; $\theta = 180^\circ$ gives $W = -Fs$.

4.2 Kinetic and Potential Energy

🕒 Estimated time: 16 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

 Quick Summary

Kinetic energy is the energy of a body due to its motion, $\frac{1}{2}mv^2$, and gravitational potential energy is the energy due to its position, mgh .

 Real-Life Example

A pile driver lifted high stores potential energy; released, it converts that into kinetic energy and drives a post into the ground.

 Detailed Explanation**Definition**

Kinetic energy is the energy possessed by a body by virtue of its motion. Potential energy is the energy possessed by a body by virtue of its position or configuration.

Key Idea

Kinetic energy depends on the square of velocity, so doubling the speed quadruples the kinetic energy. This is why stopping distances grow so steeply with speed and why speed limits matter far more than intuition suggests.

Working Principle

The work–energy theorem states that the net work done on a body equals its change in kinetic energy: $W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$. Potential energy is measured relative to a chosen reference level, usually the ground.

Applications

- A stretched bow, a compressed spring and water in a dam all store potential energy.
- Doubling a car's speed quadruples its kinetic energy and roughly quadruples its braking distance.
- Hydroelectric power converts the potential energy of stored water into electrical energy.

 Points to Remember (Exam Focus)

- ✓ Kinetic energy $KE = \frac{1}{2}mv^2$; potential energy $PE = mgh$.
- ✓ $KE \propto v^2$, so doubling speed gives four times the kinetic energy.
- ✓ Energy is a scalar; SI unit is the joule.
- ✓ Work–energy theorem: net work done = change in kinetic energy.
- ✓ Potential energy depends on the reference level chosen; only changes in it are physically meaningful.

 Common Student Mistakes

-  Writing $KE = mv^2$ and omitting the factor of $\frac{1}{2}$.
-  Assuming doubling speed doubles kinetic energy; it quadruples it.
-  Forgetting to state the reference level when calculating potential energy.

Memory Hook

'Half-em-vee-squared' for kinetic, 'em-gee-aitch' for potential — and 'speed counts twice over'.

Advanced Insights

Potential energy is a property of a system, not of a single body. The 'potential energy of a brick' is really the energy of the earth–brick system, which is why only differences in potential energy have physical meaning.

Formula Section

Kinetic Energy

Difficulty	● Easy ★☆☆
Formula	$KE = \frac{1}{2} m v^2$
Meaning of Variables	KE = kinetic energy in J, m = mass in kg, v = speed in m/s.
When to Use	For any moving body, and in every work–energy theorem problem.
Common Mistakes	Squaring the mass instead of the velocity, or dropping the $\frac{1}{2}$.
Shortcut / Memory Trick	Triple the speed → nine times the kinetic energy.

Gravitational Potential Energy

Difficulty	● Easy ★☆☆
Formula	$PE = m g h$
Meaning of Variables	PE = potential energy in J, m = mass in kg, $g = 9.8 \text{ m/s}^2$, h = height above the reference level.
When to Use	For a body raised above a chosen reference level near the earth's surface.
Common Mistakes	Using the total path length instead of the vertical height.
Shortcut / Memory Trick	Only the vertical height matters — the path taken is irrelevant.

Work–Energy Theorem

Difficulty	● Medium ★★☆☆
Formula	$W_{\text{net}} = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$
Meaning of Variables	W_{net} = net work done, u = initial speed, v = final speed, m = mass.
When to Use	To find final speed or work done without needing the acceleration or time.
Common Mistakes	Using only one force instead of the net work done by all forces.

Shortcut / Memory Trick

If the speed is unchanged, the net work done is zero.

Derivation Section **Derivation: Kinetic Energy Expression $\frac{1}{2}mv^2$** **Why This Derivation Matters**

It shows that the kinetic energy formula follows directly from the definition of work and the equations of motion, rather than being an arbitrary definition.

Assumptions Used

- A constant force acts on a body of constant mass along a straight line.
- The body starts from rest.

Step-by-Step Derivation

Step 1: Let a constant force F act on a body of mass m initially at rest, moving it a distance s and giving it a final velocity v .

Step 2: The work done by the force is $W = F \times s$.

Step 3: By Newton's second law, $F = ma$, so $W = mas$.

Step 4: From the third equation of motion with $u = 0$, $v^2 = 2as$, which gives $as = v^2/2$.

Step 5: Substituting $as = v^2/2$ into $W = mas$ gives $W = \frac{1}{2}mv^2$. Since all this work has gone into the motion of the body, its kinetic energy is $\frac{1}{2}mv^2$.

Final Result

$$KE = \frac{1}{2} m v^2$$

Exam Perspective

This derivation is commonly asked for 3 marks. Marks are given for using $F = ma$, for using the third equation of motion, and for the final substitution — show all three explicitly.

II Quick Recap — Topics Covered So Far

- **Work Done by a Force** — Work is done only when a force produces a displacement in its own direction, and it is measured as the product of the force and the displacement along the force.
- **Kinetic and Potential Energy** — Kinetic energy is the energy of a body due to its motion, $\frac{1}{2}mv^2$, and gravitational potential energy is the energy due to its position, mgh .

4.3 Conservation of Energy

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

 Quick Summary

Energy can neither be created nor destroyed, only transformed from one form to another, so the total energy of an isolated system remains constant.

Real-Life Example

A swinging pendulum trades potential energy for kinetic energy and back again, and its total stays the same until friction slowly converts it into heat.

Detailed Explanation

Definition

The law of conservation of energy states that energy can neither be created nor destroyed; it can only be transformed from one form to another, and the total energy of an isolated system remains constant.

Key Idea

For a freely falling body the sum $KE + PE$ is constant at every point of the fall. At the top it is all potential; at the bottom it is all kinetic; in between it is shared, but the total never changes.

Working Principle

Energy exists in many forms — mechanical, heat, light, sound, chemical, electrical and nuclear — and devices are simply energy converters. A bulb converts electrical energy to light and heat; a loudspeaker converts electrical to sound.

Applications

- A simple pendulum continuously converts potential energy to kinetic energy and back.
- A hydroelectric plant converts potential energy of water to electrical energy.
- Photosynthesis converts light energy into chemical energy stored in glucose.

✓ Points to Remember (Exam Focus)

- ✓ Total mechanical energy = $KE + PE$ remains constant if only gravity acts.
- ✓ For a body falling from height h : at the top $E = mgh$; at the bottom $E = \frac{1}{2}mv^2$.
- ✓ Energy is never lost — apparent losses go into heat and sound.
- ✓ Devices such as motors, generators and bulbs are energy converters.
- ✓ No machine can be 100% efficient because some energy always becomes heat.

Common Student Mistakes

-  Saying energy is 'used up' or 'lost' — it is only converted into a less useful form.
-  Applying conservation of mechanical energy where friction acts, without accounting for heat.
-  Forgetting that at the mid-point of a fall the energy is shared, not doubled.

Memory Hook

'Energy changes clothes but never leaves the party.'

Advanced Insights

Einstein's $E = mc^2$ extended conservation to include mass itself. In a nuclear reaction a tiny amount of mass converts to an enormous amount of energy, so the conserved quantity is really mass-energy taken together.

Formula Section

Conservation of Mechanical Energy

Difficulty	● Medium ★★☆☆
Formula	$mgh + \frac{1}{2}mu^2 = mgh' + \frac{1}{2}mv^2$
Meaning of Variables	h and h' = heights at the two positions, u and v = speeds at those positions, m = mass.
When to Use	For any motion under gravity alone, where friction and air resistance are negligible.
Common Mistakes	Applying it where friction is significant, which invalidates the assumption.
Shortcut / Memory Trick	For a body dropped from rest through height h , $v = \sqrt{2gh}$ — the mass cancels.

4.4 Power and Commercial Unit of Energy

 **Estimated time:** 12 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Power is the rate of doing work, measured in watts, and electrical energy in daily use is measured in kilowatt-hours, the commercial unit.

Real-Life Example

Two people carrying identical loads up the same stairs do equal work, but the one who arrives first has developed more power.

Detailed Explanation

Definition

Power is the rate of doing work or the rate of transfer of energy, $P = W/t$. Its SI unit is the watt (W), equal to one joule per second.

Key Idea

Work and power are different questions: work asks 'how much energy was transferred' and power asks 'how quickly'. A slow machine and a fast one can do exactly the same work.

Working Principle

The commercial unit of electrical energy is the kilowatt-hour, the energy consumed by a 1 kW device running for 1 hour. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$, and it is the 'unit' on an electricity bill.

Applications

- A 100 W bulb burning for 10 hours consumes 1 kWh — one unit of electricity.
- Power ratings on appliances let consumers estimate running cost.
- 1 horsepower = 746 W, still used for engine and motor ratings.

✓ Points to Remember (Exam Focus)

- ✓ $P = W/t$; SI unit is the watt, $1 \text{ W} = 1 \text{ J/s}$.
- ✓ Power can also be written $P = F \times v$ for a constant force.
- ✓ 1 kilowatt = 1000 W; 1 horsepower = 746 W.
- ✓ $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$ — the commercial unit of electrical energy.
- ✓ Energy consumed (in kWh) = power in kW $\times$ time in hours.

⚠ Common Student Mistakes

- ⚠ Treating the kilowatt-hour as a unit of power — it is a unit of energy.
- ⚠ Forgetting to convert watts to kilowatts and minutes to hours before calculating units consumed.
- ⚠ Confusing the watt (power) with the joule (energy).

**Memory Hook**

'Work is how much, Power is how fast' — and 'kWh is energy, kW is power'.

**Advanced Insights**

$P = Fv$ explains why a vehicle's top speed is set by engine power rather than force: as speed rises, air drag grows roughly as v^2 , so the power needed grows as v^3 — which is why doubling top speed demands about eight times the power.

Formula Section**Power**

Difficulty	● Easy ★☆☆
Formula	$P = W / t = F \times v$
Meaning of Variables	P = power in W, W = work in J, t = time in s, F = force in N, v = velocity in m/s.

When to Use	To find how fast work is being done, or the force a machine exerts at a known speed.
Common Mistakes	Using time in minutes without converting to seconds.
Shortcut / Memory Trick	Use $P = Fv$ when speed is constant and force is known — no need to find work first.

Commercial Unit of Energy

Difficulty	● Easy ★☆☆
Formula	$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$
Meaning of Variables	kWh = kilowatt-hour, the energy used by a 1 kW appliance in 1 hour.
When to Use	For all electricity-bill and household-consumption calculations.
Common Mistakes	Writing $1 \text{ kWh} = 3600 \text{ J}$ by forgetting the factor of 1000 for kilo.
Shortcut / Memory Trick	$\text{kWh} = (\text{watts} \div 1000) \times \text{hours}$.

II Quick Recap — Topics Covered So Far

- Conservation of Energy — Energy can neither be created nor destroyed, only transformed from one form to another, so the total energy of an isolated system remains constant.
- Power and Commercial Unit of Energy — Power is the rate of doing work, measured in watts, and electrical energy in daily use is measured in kilowatt-hours, the commercial unit.

Chapter 4 — End of Chapter Revision

All Memory Hooks

- Work Done by a Force: 'No displacement, no work' — and 'perpendicular force does nothing at all'.
- Kinetic and Potential Energy: 'Half-em-vee-squared' for kinetic, 'em-gee-aitch' for potential — and 'speed counts twice over'.
- Conservation of Energy: 'Energy changes clothes but never leaves the party.'
- Power and Commercial Unit of Energy: 'Work is how much, Power is how fast' — and 'kWh is energy, kW is power'.

All Common Mistakes

-  Saying work is done when a body is merely held stationary — displacement is required.
-  Omitting $\cos \theta$ when the force acts at an angle to the displacement.
-  Treating work as a vector; it is a scalar.
-  Writing $KE = mv^2$ and omitting the factor of $\frac{1}{2}$.
-  Assuming doubling speed doubles kinetic energy; it quadruples it.
-  Forgetting to state the reference level when calculating potential energy.
-  Saying energy is 'used up' or 'lost' — it is only converted into a less useful form.
-  Applying conservation of mechanical energy where friction acts, without accounting for heat.
-  Forgetting that at the mid-point of a fall the energy is shared, not doubled.
-  Treating the kilowatt-hour as a unit of power — it is a unit of energy.
-  Forgetting to convert watts to kilowatts and minutes to hours before calculating units consumed.
-  Confusing the watt (power) with the joule (energy).

Formula Sheet

Formula	Meaning / Use
Work Done: $W = F s \cos \theta$	For any constant force producing a displacement, including forces at an angle.
Kinetic Energy: $KE = \frac{1}{2} m v^2$	For any moving body, and in every work–energy theorem problem.
Gravitational Potential Energy: $PE = m g h$	For a body raised above a chosen reference level near the earth's surface.
Work–Energy Theorem: $W_{\text{net}} = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$	To find final speed or work done without needing the acceleration or time.

Conservation of Mechanical Energy: $mgh + \frac{1}{2}mu^2 = mgh' + \frac{1}{2}mv^2$	For any motion under gravity alone, where friction and air resistance are negligible.
Power: $P = W / t = F \times v$	To find how fast work is being done, or the force a machine exerts at a known speed.
Commercial Unit of Energy: $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$	For all electricity-bill and household-consumption calculations.

Quick Revision Section

- **Work Done by a Force:** Work is done only when a force produces a displacement in its own direction, and it is measured as the product of the force and the displacement along the force.
- **Kinetic and Potential Energy:** Kinetic energy is the energy of a body due to its motion, $\frac{1}{2}mv^2$, and gravitational potential energy is the energy due to its position, mgh .
- **Conservation of Energy:** Energy can neither be created nor destroyed, only transformed from one form to another, so the total energy of an isolated system remains constant.
- **Power and Commercial Unit of Energy:** Power is the rate of doing work, measured in watts, and electrical energy in daily use is measured in kilowatt-hours, the commercial unit.

Chapter 5: Sound

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea of vibration and oscillation.
- Understanding that matter exists as solid, liquid and gas.
- Basic knowledge of speed, distance and time.
- Familiarity with the words echo and reflection.

5.1 Production and Propagation of Sound

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Sound is produced by vibrating bodies and travels as a longitudinal wave through a material medium; it cannot travel through vacuum.

Real-Life Example

An electric bell ringing inside a bell jar falls silent as the air is pumped out, even though it can still be seen vibrating.

Detailed Explanation

Definition

Sound is a form of energy produced by vibrating bodies that travels through a material medium as a longitudinal wave and produces the sensation of hearing.

Key Idea

In a longitudinal wave the particles of the medium vibrate along the direction of propagation, creating alternate regions of compression (high pressure, particles close together) and rarefaction (low pressure, particles far apart).

Working Principle

The particles of the medium do not travel with the wave — they only oscillate about their mean positions. What travels is the disturbance and the energy, not the matter.

Applications

- A vibrating tuning fork, a plucked string and a beating drum all produce sound.
- Sound needs a medium — astronauts in space communicate by radio, not by shouting.
- The larynx produces speech by vibrating vocal cords.

✓ Points to Remember (Exam Focus)

- ✓ Sound is a longitudinal wave; particles vibrate parallel to the direction of propagation.
- ✓ Compressions are regions of high pressure and density; rarefactions are low pressure and density.
- ✓ Sound requires a material medium and cannot travel through vacuum.
- ✓ The particles of the medium do not move along with the wave.
- ✓ Sound travels fastest in solids, slower in liquids and slowest in gases.

⚠ Common Student Mistakes

- ⚠ Calling sound a transverse wave — it is longitudinal in all media except in some solids.
- ⚠ Thinking air molecules travel from the source to the ear.
- ⚠ Assuming sound travels faster in gases because gases are lighter; it travels fastest in solids.



Memory Hook

'Compression = Crowded, Rarefaction = Rare' — and 'Solids Speed sound, Vacuum Silences it'.



Advanced Insights

Sound travels faster in solids because the elastic modulus rises far more steeply than the density between gases and solids, and the speed depends on the ratio of the two. Steel carries sound about fifteen times faster than air.

5.2 Characteristics of a Sound Wave

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

A sound wave is described by its frequency, wavelength, amplitude, time period and speed, and these determine the pitch, loudness and quality of the sound heard.



Real-Life Example

A mosquito's whine is high-pitched because of its high frequency, while a lion's roar is loud because of its large amplitude — two different properties entirely.



Detailed Explanation

Definition

Frequency is the number of complete oscillations per second, measured in hertz. Wavelength is the distance between two consecutive compressions. Amplitude is the maximum displacement of a particle from its mean position. Time period is the time for one complete oscillation.

Key Idea

Pitch depends on frequency, loudness depends on amplitude, and timbre or quality distinguishes two sounds of the same pitch and loudness — which is why a violin and a flute playing the same note sound different.

Working Principle

The wave equation $v = f\lambda$ links speed, frequency and wavelength. When sound passes from one medium to another its frequency stays the same, so a change of speed forces a change of wavelength.

Applications

- Audible range for humans: 20 Hz to 20,000 Hz.
- Infrasound is below 20 Hz (produced by elephants, earthquakes); ultrasound is above 20,000 Hz (bats, dolphins).
- Loudness is measured in decibels; above about 80 dB sound becomes physically harmful.

✓ Points to Remember (Exam Focus)

- ✓ $v = f\lambda$; frequency $f = 1/T$ where T is the time period.
- ✓ Pitch $\leftrightarrow$ frequency; Loudness $\leftrightarrow$ amplitude; Quality $\leftrightarrow$ waveform.
- ✓ Audible range: 20 Hz to 20 kHz.
- ✓ Frequency does NOT change when sound enters a new medium — speed and wavelength do.
- ✓ Speed of sound in air at 20 °C is about 344 m/s and increases with temperature.

⚠ Common Student Mistakes

- ⚠ Saying a louder sound has a higher pitch — loudness and pitch are independent.
- ⚠ Assuming the frequency changes when sound passes from air into water.
- ⚠ Confusing amplitude (displacement) with wavelength (distance between compressions).

🧠 Memory Hook

'Frequency = Fine or Fat note (pitch); Amplitude = Amount of loudness.'

🚀 Advanced Insights

Because the speed of sound in air rises with the square root of absolute temperature, sound travels faster on a hot day. The resulting refraction is why distant sounds carry unusually far across a lake on a cold morning.

Formula Section

Wave Equation

Difficulty	● Easy ★☆☆
Formula	$v = f \lambda$
Meaning of Variables	v = speed of the wave in m/s, f = frequency in Hz, λ = wavelength in m.
When to Use	Whenever two of speed, frequency and wavelength are known and the third is required.
Common Mistakes	Mixing units — wavelength in cm with speed in m/s.
Shortcut / Memory Trick	Frequency stays constant across media; use it to link the wavelengths in two media.

Frequency and Time Period

Difficulty	● Easy ★☆☆
Formula	$f = 1 / T$
Meaning of Variables	f = frequency in Hz, T = time period in seconds.
When to Use	To convert between the two descriptions of how fast a wave oscillates.
Common Mistakes	Treating them as directly rather than inversely proportional.
Shortcut / Memory Trick	Higher frequency always means a shorter time period.

II Quick Recap — Topics Covered So Far

- Production and Propagation of Sound — Sound is produced by vibrating bodies and travels as a longitudinal wave through a material medium; it cannot travel through vacuum.
- Characteristics of a Sound Wave — A sound wave is described by its frequency, wavelength, amplitude, time period and speed, and these determine the pitch, loudness and quality of the sound heard.

5.3 Reflection of Sound — Echo and Reverberation

 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Sound obeys the laws of reflection; a distinctly repeated sound is an echo, while the persistence of sound due to repeated reflection in a hall is reverberation.

Real-Life Example

Shouting into a well returns a clear echo, but the same shout in a large empty hall becomes a lingering blur — the difference is the time gap between reflections.

Detailed Explanation

Definition

An echo is a sound heard after reflection from a distant obstacle, distinct from the original sound. Reverberation is the persistence of sound in a large hall due to repeated reflections.

Key Idea

The sensation of a sound persists in the human brain for about 0.1 s. For an echo to be heard separately, the reflected sound must arrive at least 0.1 s after the original — which fixes a minimum distance of about 17.2 m.

Working Principle

In 0.1 s sound in air travels $344 \times 0.1 = 34.4$ m. Since this is the total to-and-fro path, the reflecting surface must be at least 17.2 m away.

Applications

- Megaphones and horns use multiple reflections to send sound in one direction.
- Stethoscopes carry sound to the doctor's ears by repeated reflection inside the tube.
- Concert halls have curved ceilings and sound-absorbing curtains to control reverberation.

✓ Points to Remember (Exam Focus)

- ✓ Laws of reflection apply to sound: angle of incidence = angle of reflection, all in one plane.
- ✓ Persistence of hearing is about 0.1 s.
- ✓ Minimum distance for an echo in air is 17.2 m (at 344 m/s).
- ✓ Reverberation is excessive in large halls and is reduced by absorbent materials.
- ✓ Reflecting surfaces for sound must be large compared with the wavelength.

⚠ Common Student Mistakes

- ⚠ Using 34.4 m as the minimum echo distance, forgetting that sound travels both to and from the obstacle.
- ⚠ Confusing echo (distinct repetition) with reverberation (continuous persistence).
- ⚠ Forgetting to halve the total path when calculating the distance of a reflector.



Memory Hook

'Echo needs 17.2 m — half of 34.4, because sound goes there AND back.'

Advanced Insights

Sabine's formula relates reverberation time to a hall's volume and total absorption, and it made acoustics an engineering discipline. Boston Symphony Hall, designed with it in 1900, is still ranked among the finest concert halls in the world.

Formula Section

Distance of a Reflecting Surface

Difficulty	● Medium ★★☆☆
Formula	$d = v \times t / 2$
Meaning of Variables	d = distance of the obstacle, v = speed of sound, t = total time between producing the sound and hearing the echo.
When to Use	For echo, sonar and depth-sounding problems.
Common Mistakes	Omitting the division by 2, which doubles the answer.
Shortcut / Memory Trick	The sound covers the distance twice, so always halve.

5.4 Ultrasound and Its Applications

 **Estimated time:** 12 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Ultrasound is sound above 20,000 Hz; its short wavelength lets it travel in narrow beams and be reflected from small objects, which makes it valuable in medicine and industry.

Real-Life Example

A prenatal scan, a ship's depth sounder and a bat navigating a dark cave all rely on exactly the same physics.

Detailed Explanation

Definition

Ultrasound is sound of frequency higher than 20 kHz, beyond the upper limit of human hearing. SONAR stands for Sound Navigation And Ranging and uses ultrasound to find the distance and direction of underwater objects.

Key Idea

High frequency means short wavelength, and a wave reflects well only from objects larger than its wavelength. Ultrasound's short wavelength is therefore what lets it detect small flaws and image small structures.

Working Principle

In SONAR a transmitter emits ultrasonic pulses that reflect from the sea bed or an object; the time delay gives the distance as $d = vt/2$. Echolocation in bats and dolphins works by the same principle.

Applications

- Medical: imaging the foetus, kidneys and liver; breaking kidney stones into fine grains.
- Industrial: detecting cracks in metal castings; cleaning spiral tubes and delicate parts.
- Navigation: measuring sea depth, locating shoals of fish, submarines and shipwrecks.

✓ Points to Remember (Exam Focus)

- ✓ Ultrasound: frequency above 20,000 Hz; Infrasound: below 20 Hz.
- ✓ SONAR uses the formula $d = vt/2$, the same as for echoes.
- ✓ Speed of sound in sea water is about 1500 m/s, far higher than in air.
- ✓ Bats, dolphins and porpoises navigate by echolocation.
- ✓ Ultrasound is preferred for medical imaging because it is non-ionising and safe.

⚠ Common Student Mistakes

- ⚠ Confusing ultrasound (above hearing range) with infrasound (below hearing range).
- ⚠ Using the speed of sound in air for a SONAR problem set in water.
- ⚠ Forgetting to halve the time in echo-ranging calculations.



Memory Hook

'ULTRA = Up above 20 kHz; INFRA = In-ferior, below 20 Hz.'



Advanced Insights

Resolution in ultrasound imaging is limited to about one wavelength, so higher frequencies give sharper images but penetrate less deeply. Sonographers trade resolution against depth by choosing the probe frequency — a fundamental limit, not a technological one.

Formula Section



SONAR Range Equation

Difficulty	● Medium ★★☆☆
Formula	$d = v \times t / 2$
Meaning of Variables	d = depth or distance of object, v = speed of sound in the medium, t = total time for the pulse to return.
When to Use	For sea-depth measurement, submarine detection and any echo-ranging problem.

Common Mistakes	Using 344 m/s in water instead of about 1500 m/s.
Shortcut / Memory Trick	Same formula as an echo — always divide by 2 for the return trip.

II Quick Recap — Topics Covered So Far

- Reflection of Sound — Echo and Reverberation — Sound obeys the laws of reflection; a distinctly repeated sound is an echo, while the persistence of sound due to repeated reflection in a hall is reverberation.
- Ultrasound and Its Applications — Ultrasound is sound above 20,000 Hz; its short wavelength lets it travel in narrow beams and be reflected from small objects, which makes it valuable in medicine and industry.

Chapter 5 — End of Chapter Revision

All Memory Hooks

- Production and Propagation of Sound: 'Compression = Crowded, Rarefaction = Rare' — and 'Solids Speed sound, Vacuum Silences it'.
- Characteristics of a Sound Wave: 'Frequency = Fine or Fat note (pitch); Amplitude = Amount of loudness.'
- Reflection of Sound — Echo and Reverberation: 'Echo needs 17.2 m — half of 34.4, because sound goes there AND back.'
- Ultrasound and Its Applications: 'ULTRA = Up above 20 kHz; INFRA = In-ferior, below 20 Hz.'

All Common Mistakes

-  Calling sound a transverse wave — it is longitudinal in all media except in some solids.
-  Thinking air molecules travel from the source to the ear.
-  Assuming sound travels faster in gases because gases are lighter; it travels fastest in solids.
-  Saying a louder sound has a higher pitch — loudness and pitch are independent.
-  Assuming the frequency changes when sound passes from air into water.
-  Confusing amplitude (displacement) with wavelength (distance between compressions).
-  Using 34.4 m as the minimum echo distance, forgetting that sound travels both to and from the obstacle.
-  Confusing echo (distinct repetition) with reverberation (continuous persistence).
-  Forgetting to halve the total path when calculating the distance of a reflector.
-  Confusing ultrasound (above hearing range) with infrasound (below hearing range).
-  Using the speed of sound in air for a SONAR problem set in water.
-  Forgetting to halve the time in echo-ranging calculations.

Formula Sheet

Formula	Meaning / Use
Wave Equation: $v = f \lambda$	Whenever two of speed, frequency and wavelength are known and the third is required.
Frequency and Time Period: $f = 1 / T$	To convert between the two descriptions of how fast a wave oscillates.
Distance of a Reflecting Surface: $d = v \times t / 2$	For echo, sonar and depth-sounding problems.

SONAR Range Equation: $d = v \times t / 2$

For sea-depth measurement, submarine detection and any echo-ranging problem.

Quick Revision Section

- **Production and Propagation of Sound:** Sound is produced by vibrating bodies and travels as a longitudinal wave through a material medium; it cannot travel through vacuum.
- **Characteristics of a Sound Wave:** A sound wave is described by its frequency, wavelength, amplitude, time period and speed, and these determine the pitch, loudness and quality of the sound heard.
- **Reflection of Sound — Echo and Reverberation:** Sound obeys the laws of reflection; a distinctly repeated sound is an echo, while the persistence of sound due to repeated reflection in a hall is reverberation.
- **Ultrasound and Its Applications:** Ultrasound is sound above 20,000 Hz; its short wavelength lets it travel in narrow beams and be reflected from small objects, which makes it valuable in medicine and industry.

Chapter 6: Units and Measurements

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Familiarity with metre, kilogram and second.
- Ability to work with powers of ten.
- Basic idea of area and volume.
- Knowledge that every measurement carries some uncertainty.

6.1 Physical Quantities and the SI System

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Physical quantities are measured against agreed standards, and the SI system defines seven base units from which all other units are derived.

Real-Life Example

The Mars Climate Orbiter was lost in 1999 because one team used pounds and another newtons — a \$125 million reminder that units are not a formality.

Detailed Explanation

Definition

A physical quantity is anything that can be measured. A unit is the agreed standard against which it is measured. Base units are independent and cannot be derived from one another; derived units are combinations of base units.

Key Idea

There are exactly seven SI base units. Every other unit in physics — newton, joule, watt, pascal — is built from these seven by multiplication and division alone.

Working Principle

A measurement is meaningless without a unit, and the numerical value is inversely proportional to the size of the unit chosen. The same length is 1 m or 100 cm — a larger unit gives a smaller number.

Applications

- Newton (N) = $\text{kg}\cdot\text{m}/\text{s}^2$; Joule (J) = $\text{kg}\cdot\text{m}^2/\text{s}^2$; Watt (W) = $\text{kg}\cdot\text{m}^2/\text{s}^3$.
- Prefixes: kilo (10^3), mega (10^6), giga (10^9), milli (10^{-3}), micro (10^{-6}), nano (10^{-9}).
- Astronomical unit and light year are convenient units for very large distances.

✓ Points to Remember (Exam Focus)

- ✓ Seven SI base units: metre (length), kilogram (mass), second (time), ampere (current), kelvin (temperature), mole (amount of substance), candela (luminous intensity).
- ✓ Derived units are combinations of base units; e.g. $1 \text{ N} = 1 \text{ kg}\cdot\text{m}/\text{s}^2$.
- ✓ Unit symbols named after scientists are capitalised (N, J, W, K) but written in full in lower case (newton, joule).
- ✓ Never write a plural 's' on a unit symbol — 5 kg, not 5 kgs.
- ✓ 1 light year = $9.46 \times 10^{15} \text{ m}$ — a unit of distance, not of time.

⚠ Common Student Mistakes

- ⚠ Calling the light year a unit of time; it is a unit of distance.
- ⚠ Writing 'Kg' or 'KG' instead of 'kg' — only the prefix rules decide capitalisation.
- ⚠ Treating the newton as a base unit; it is derived.



Memory Hook

'Metre, Kilogram, Second, Ampere, Kelvin, Mole, Candela' — 'My King Sends A Knight, Mounted, Charging.'



Advanced Insights

Since 2019 every SI base unit is defined by fixing a fundamental constant — the kilogram through Planck's constant, the metre through the speed of light. No physical artefact remains, so the units cannot drift over time.

6.2 Measuring Instruments — Vernier Callipers and Screw Gauge

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

A vernier calliper measures to 0.1 mm and a screw gauge to 0.01 mm, using a scale that magnifies a fraction of the smallest main-scale division.



Real-Life Example

A mechanic checks the diameter of a piston with a screw gauge because a metre rule simply cannot resolve the hundredth of a millimetre that decides whether an engine seizes.

Detailed Explanation

Definition

Least count is the smallest measurement that an instrument can read accurately. For a vernier calliper it is the difference between one main-scale division and one vernier-scale division.

Key Idea

Both instruments work by the same trick: a second scale is made slightly different from the first, so the point at which two divisions coincide reveals a fraction of the smallest main-scale division.

Working Principle

Total reading = main scale reading + (vernier or circular scale coincidence $\times$ least count). Zero error must be found first and subtracted algebraically from every reading.

Applications

- Vernier calliper: internal and external diameters, depths of hollow cylinders.
- Screw gauge: thickness of a wire, a sheet of paper or a glass slide.
- Both are used in workshops, laboratories and quality control.

✓ Points to Remember (Exam Focus)

- ✓ Least count of vernier calliper = $1 \text{ MSD} - 1 \text{ VSD}$, usually 0.1 mm or 0.01 cm.
- ✓ Least count of screw gauge = pitch / number of circular scale divisions, usually 0.01 mm.
- ✓ Pitch = distance moved on the main scale in one complete rotation of the screw.
- ✓ Total reading = MSR + (coincidence $\times$ LC), then correct for zero error.
- ✓ Positive zero error is subtracted; negative zero error is added.

⚠ Common Student Mistakes

- ⚠ Adding a positive zero error instead of subtracting it.
- ⚠ Using the number of vernier divisions instead of the coincidence number in the formula.
- ⚠ Forgetting to convert the final reading into the required unit.



Memory Hook

'Correct reading = Observed reading – Zero error' — sign and all, every single time.



Advanced Insights

The vernier principle dates to 1631 and remains in use because it needs no power and cannot drift. Digital callipers replace the eye with a capacitive encoder but rely on exactly the same geometrical idea.

Formula Section

Least Count of a Vernier Calliper

Difficulty	● Medium ★★☆☆
Formula	$LC = 1 \text{ MSD} - 1 \text{ VSD}$
Meaning of Variables	MSD = one main scale division, VSD = one vernier scale division.
When to Use	Before taking any reading with a vernier calliper.
Common Mistakes	Dividing instead of subtracting.
Shortcut / Memory Trick	LC also equals (value of 1 MSD) $\div$ (number of vernier divisions).

Least Count of a Screw Gauge

Difficulty	● Medium ★★☆☆
Formula	$LC = \text{Pitch} / \text{Number of circular scale divisions}$
Meaning of Variables	Pitch = distance advanced in one full rotation, usually 0.5 mm or 1 mm.
When to Use	Before taking any reading with a screw gauge or micrometer.
Common Mistakes	Using the total number of main scale divisions instead of circular scale divisions.
Shortcut / Memory Trick	Pitch 1 mm with 100 divisions gives the standard LC of 0.01 mm.

II Quick Recap — Topics Covered So Far

- Physical Quantities and the SI System — Physical quantities are measured against agreed standards, and the SI system defines seven base units from which all other units are derived.
- Measuring Instruments — Vernier Callipers and Screw Gauge — A vernier calliper measures to 0.1 mm and a screw gauge to 0.01 mm, using a scale that magnifies a fraction of the smallest main-scale division.

6.3 Errors and Significant Figures

 **Estimated time:** 13 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Every measurement carries an error; significant figures record how precisely a quantity is known, and a calculated result can never be more precise than the least precise measurement used.

Real-Life Example

Reporting a body's mass as 52.3846 kg after weighing it on a bathroom scale claims a precision the instrument never had.

Detailed Explanation

Definition

Systematic errors have a definite cause and always shift readings in the same direction; random errors vary unpredictably. Significant figures are the digits in a measurement that are known reliably plus one uncertain digit.

Key Idea

Accuracy is closeness to the true value; precision is the closeness of repeated measurements to one another. A measurement can be precise but inaccurate — for example, a scale with a zero error.

Working Principle

Random errors are reduced by taking the mean of many readings. Systematic errors are not — they must be removed by correcting the instrument, which is why zero error is checked before measuring.

Applications

- In addition and subtraction, the result keeps as many decimal places as the least precise term.
- In multiplication and division, the result keeps as many significant figures as the least precise factor.
- Zero error in callipers and screw gauges is a classic systematic error.

✓ Points to Remember (Exam Focus)

- ✓ All non-zero digits are significant; zeros between non-zero digits are significant.
- ✓ Leading zeros are not significant; trailing zeros after a decimal point are.
- ✓ Accuracy = closeness to the true value; Precision = repeatability.
- ✓ Random errors are reduced by averaging; systematic errors are not.
- ✓ The final answer must not have more significant figures than the least precise input.

⚠ Common Student Mistakes

- ⚠ Quoting a calculator result to eight digits when the data justify only three.
- ⚠ Counting leading zeros in 0.0025 as significant — there are only two significant figures.
- ⚠ Assuming averaging removes all errors; it only reduces random ones.



Memory Hook

'Accurate = on target; Precise = tightly grouped' — you can be one without the other.

 Advanced Insights

Errors propagate predictably: for a product or quotient, the fractional errors add. So if a radius is known to 1%, the volume of a sphere is known only to 3%, because the radius is cubed.

II Quick Recap — Topics Covered So Far

- **Errors and Significant Figures** — Every measurement carries an error; significant figures record how precisely a quantity is known, and a calculated result can never be more precise than the least precise measurement used.

Chapter 6 — End of Chapter Revision

All Memory Hooks

- Physical Quantities and the SI System: 'Metre, Kilogram, Second, Ampere, Kelvin, Mole, Candela' — 'My King Sends A Knight, Mounted, Charging.'
- Measuring Instruments — Vernier Callipers and Screw Gauge: 'Correct reading = Observed reading – Zero error' — sign and all, every single time.
- Errors and Significant Figures: 'Accurate = on target; Precise = tightly grouped' — you can be one without the other.

All Common Mistakes

-  Calling the light year a unit of time; it is a unit of distance.
-  Writing 'Kg' or 'KG' instead of 'kg' — only the prefix rules decide capitalisation.
-  Treating the newton as a base unit; it is derived.
-  Adding a positive zero error instead of subtracting it.
-  Using the number of vernier divisions instead of the coincidence number in the formula.
-  Forgetting to convert the final reading into the required unit.
-  Quoting a calculator result to eight digits when the data justify only three.
-  Counting leading zeros in 0.0025 as significant — there are only two significant figures.
-  Assuming averaging removes all errors; it only reduces random ones.

Formula Sheet

Formula	Meaning / Use
Least Count of a Vernier Calliper: $LC = 1 \text{ MSD} - 1 \text{ VSD}$	Before taking any reading with a vernier calliper.
Least Count of a Screw Gauge: $LC = \text{Pitch} / \text{Number of circular scale divisions}$	Before taking any reading with a screw gauge or micrometer.

Quick Revision Section

- Physical Quantities and the SI System: Physical quantities are measured against agreed standards, and the SI system defines seven base units from which all other units are derived.

- Measuring Instruments — Vernier Callipers and Screw Gauge: A vernier calliper measures to 0.1 mm and a screw gauge to 0.01 mm, using a scale that magnifies a fraction of the smallest main-scale division.
- Errors and Significant Figures: Every measurement carries an error; significant figures record how precisely a quantity is known, and a calculated result can never be more precise than the least precise measurement used.

Chapter 7: Centre of Mass and Equilibrium of Rigid Bodies

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Newton's laws of motion and the idea of balanced forces.
- Concept of weight acting downward on a body.
- Ability to resolve forces into components.
- Basic idea of the turning effect of a force.

7.1 Centre of Mass and Centre of Gravity

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**    

Quick Summary

The centre of mass is the point at which the whole mass of a body may be considered concentrated, and the centre of gravity is the point through which its entire weight acts.

Real-Life Example

A high jumper using the Fosbury flop arches so much that their centre of mass passes below the bar even as their body passes above it.

Detailed Explanation

Definition

The centre of mass of a system is the point where the entire mass may be assumed to be concentrated for the purpose of describing translational motion. The centre of gravity is the point through which the resultant weight of the body acts.

Key Idea

For bodies small compared with the earth, where g is effectively uniform, the centre of mass and the centre of gravity coincide. They separate only for very large bodies such as a mountain or a tall building.

Working Principle

For a uniform body of regular shape the centre of mass lies at its geometric centre. It need not lie within the material of the body at all — for a ring or a hollow sphere it lies in the empty space at the centre.

Applications

- The centre of mass of a uniform rod is at its mid-point; of a triangular lamina, at the centroid.
- A body thrown in the air rotates about its centre of mass while the centre of mass follows a smooth parabola.
- Vehicles are designed with a low centre of gravity to resist toppling.

✓ Points to Remember (Exam Focus)

- ✓ Centre of mass may lie outside the material of the body (ring, horseshoe, boomerang).
- ✓ Centre of mass and centre of gravity coincide when g is uniform over the body.
- ✓ For a uniform regular body, the centre of mass is at the geometric centre.
- ✓ The centre of mass of a two-particle system lies on the line joining them, closer to the heavier one.
- ✓ External forces determine the motion of the centre of mass; internal forces cannot change it.

⚠ Common Student Mistakes

- ⚠ Assuming the centre of mass must lie inside the body.
- ⚠ Treating centre of mass and centre of gravity as always identical without stating the condition.
- ⚠ Placing the centre of mass of a two-particle system midway regardless of the masses.



Memory Hook

'Centre of Mass = the balance point; heavier side pulls it closer.'



Advanced Insights

When a firework explodes in mid-air, the fragments fly apart but the centre of mass continues along the original parabola. Internal forces, however violent, cannot alter the motion of the centre of mass.

Formula Section



Centre of Mass of a Two-Particle System

Difficulty	● Medium ★★☆
Formula	$x_{cm} = (m_1x_1 + m_2x_2) / (m_1 + m_2)$
Meaning of Variables	m_1, m_2 = the two masses; x_1, x_2 = their positions from a chosen origin.
When to Use	To locate the balance point of a system of two or more particles.
Common Mistakes	Forgetting to divide by the total mass, or measuring positions from different origins.
Shortcut / Memory Trick	For equal masses the centre of mass is exactly midway.

7.2 Moment of a Force and the Principle of Moments

🕒 Estimated time: 15 mins Difficulty: ● Medium ★★☆ Exam Importance: ★★★★★

Quick Summary

The moment of a force, or torque, is the product of the force and the perpendicular distance from the pivot, and a body is in rotational equilibrium when clockwise and anticlockwise moments balance.

Real-Life Example

A door handle is fitted far from the hinges because the same push produces a much greater turning effect at a greater distance.

Detailed Explanation

Definition

The moment of a force about a point is the product of the force and the perpendicular distance of its line of action from that point. Its SI unit is the newton metre (N·m).

Key Idea

It is the perpendicular distance that matters, not the distance along the force. A force whose line of action passes through the pivot has zero moment however large it is.

Working Principle

The principle of moments states that for a body in rotational equilibrium, the sum of clockwise moments about any point equals the sum of anticlockwise moments about the same point.

Applications

- A see-saw balances when $m_1gd_1 = m_2gd_2$ about the pivot.
- Spanners are made long to increase the moment for the same applied force.
- A steelyard balance and a common beam balance both apply the principle of moments.

✓ Points to Remember (Exam Focus)

- ✓ Moment = Force $\times$ perpendicular distance from the pivot; unit N·m.
- ✓ Moment is zero if the line of action passes through the pivot.
- ✓ Principle of moments: sum of clockwise moments = sum of anticlockwise moments.
- ✓ A couple is a pair of equal, opposite, parallel forces that produce rotation without translation.
- ✓ For complete equilibrium both the net force AND the net moment must be zero.

Common Student Mistakes

-  Using the slant distance instead of the perpendicular distance to the line of action.
-  Checking only force balance and forgetting that moments must balance too.
-  Taking moments about different points for the clockwise and anticlockwise sides.

Memory Hook

'Moment = Force $\times$ Perpendicular distance' — perpendicular, always.

Advanced Insights

Torque and force are related as angular quantities to linear ones: torque plays the role of force, moment of inertia the role of mass, and angular acceleration the role of linear acceleration, giving $\tau = I\alpha$ as the rotational form of $F = ma$.

Formula Section

Moment of a Force

Difficulty	● Easy ★☆☆
Formula	$\tau = F \times d$
Meaning of Variables	τ = moment or torque in N·m, F = force in N, d = perpendicular distance from the pivot in m.
When to Use	For levers, spanners, doors, see-saws and any turning problem.
Common Mistakes	Measuring d along the force instead of perpendicular to it.
Shortcut / Memory Trick	Double the distance $\rightarrow$ double the turning effect for the same force.

Principle of Moments

Difficulty	● Medium ★★☆☆
Formula	$F_1 d_1 = F_2 d_2$
Meaning of Variables	F_1, F_2 = forces on either side of the pivot; d_1, d_2 = their perpendicular distances from it.
When to Use	For a balanced beam, see-saw or lever in rotational equilibrium.
Common Mistakes	Ignoring the weight of the beam itself when it is not uniform or not pivoted at its centre.
Shortcut / Memory Trick	Take moments about the pivot to eliminate the unknown reaction there.

II Quick Recap — Topics Covered So Far

- Centre of Mass and Centre of Gravity — The centre of mass is the point at which the whole mass of a body may be considered concentrated, and the centre of gravity is the point through which its entire weight acts.

- **Moment of a Force and the Principle of Moments** — The moment of a force, or torque, is the product of the force and the perpendicular distance from the pivot, and a body is in rotational equilibrium when clockwise and anticlockwise moments balance.

7.3 Equilibrium and Stability

Estimated time: 13 mins **Difficulty:** Medium **Exam Importance:** ★★☆☆

Quick Summary

A body is in equilibrium when both net force and net moment are zero, and its stability depends on how the centre of gravity moves when it is tilted.

Real-Life Example

A racing car with a wide track and a low body is hard to overturn, while a double-decker bus with passengers on the upper deck is much easier.

Detailed Explanation

Definition

A body is in equilibrium when the net force and the net moment acting on it are both zero. Stable equilibrium is a state in which a body returns to its original position after a small displacement.

Key Idea

A body topples when the vertical line through its centre of gravity falls outside its base. Stability is therefore improved by lowering the centre of gravity and widening the base.

Working Principle

In stable equilibrium the centre of gravity rises when the body is tilted, so it returns. In unstable equilibrium it falls, so the body topples further. In neutral equilibrium the height is unchanged, so it stays put.

Applications

- Racing cars and buses are built with a low centre of gravity and a wide wheelbase.
- A cone resting on its base is stable; balanced on its tip it is unstable; on its side it is neutral.
- A Bunsen burner has a heavy base for exactly this reason.

✓ **Points to Remember (Exam Focus)**

- ✓ Conditions for equilibrium: net force = 0 AND net moment = 0.
- ✓ Stable: centre of gravity rises on tilting, body returns.
- ✓ Unstable: centre of gravity falls on tilting, body topples.
- ✓ Neutral: centre of gravity stays at the same height, body remains where placed.
- ✓ A body topples when the vertical through its centre of gravity falls outside its base.

Common Student Mistakes

-  Checking only force balance and calling that equilibrium — a couple gives zero net force but rotation.
-  Thinking a heavy body is automatically stable; it is the position of the centre of gravity that matters.
-  Confusing neutral equilibrium with no equilibrium at all.



Memory Hook

'Low centre, Wide base = hard to topple' — and 'stable if the CG rises when you tilt it'.



Advanced Insights

A ship's stability is measured by its metacentric height. When a ship heels, buoyancy shifts sideways and creates a righting moment — which is why cargo is stowed low and why a shifting load is so dangerous.

II Quick Recap — Topics Covered So Far

- **Equilibrium and Stability** — A body is in equilibrium when both net force and net moment are zero, and its stability depends on how the centre of gravity moves when it is tilted.

Chapter 7 — End of Chapter Revision

All Memory Hooks

- Centre of Mass and Centre of Gravity: 'Centre of Mass = the balance point; heavier side pulls it closer.'
- Moment of a Force and the Principle of Moments: 'Moment = Force $\times$ Perpendicular distance' — perpendicular, always.
- Equilibrium and Stability: 'Low centre, Wide base = hard to topple' — and 'stable if the CG rises when you tilt it'.

All Common Mistakes

-  Assuming the centre of mass must lie inside the body.
-  Treating centre of mass and centre of gravity as always identical without stating the condition.
-  Placing the centre of mass of a two-particle system midway regardless of the masses.
-  Using the slant distance instead of the perpendicular distance to the line of action.
-  Checking only force balance and forgetting that moments must balance too.
-  Taking moments about different points for the clockwise and anticlockwise sides.
-  Checking only force balance and calling that equilibrium — a couple gives zero net force but rotation.
-  Thinking a heavy body is automatically stable; it is the position of the centre of gravity that matters.
-  Confusing neutral equilibrium with no equilibrium at all.

Formula Sheet

Formula	Meaning / Use
Centre of Mass of a Two-Particle System: $x_{cm} = (m_1x_1 + m_2x_2) / (m_1 + m_2)$	To locate the balance point of a system of two or more particles.
Moment of a Force: $\tau = F \times d$	For levers, spanners, doors, see-saws and any turning problem.
Principle of Moments: $F_1 d_1 = F_2 d_2$	For a balanced beam, see-saw or lever in rotational equilibrium.

Quick Revision Section

- **Centre of Mass and Centre of Gravity:** The centre of mass is the point at which the whole mass of a body may be considered concentrated, and the centre of gravity is the point through which its entire weight acts.
- **Moment of a Force and the Principle of Moments:** The moment of a force, or torque, is the product of the force and the perpendicular distance from the pivot, and a body is in rotational equilibrium when clockwise and anticlockwise moments balance.
- **Equilibrium and Stability:** A body is in equilibrium when both net force and net moment are zero, and its stability depends on how the centre of gravity moves when it is tilted.

Chapter 8: Mechanical Properties of Solids and Fluids

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Concept of force, pressure and density.
- Archimedes' principle and buoyancy from the Gravitation chapter.
- Idea that solids resist a change in shape.
- Basic understanding of area and volume.

8.1 Elasticity, Stress and Strain

 **Estimated time:** 15 mins **Difficulty:**  Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Elasticity is the property by which a body regains its original shape after the deforming force is removed, and Hooke's law states that stress is proportional to strain within the elastic limit.

Real-Life Example

A rubber band returns to shape after a gentle stretch, but pull it too far and it stays permanently longer — the elastic limit has been crossed.

Detailed Explanation

Definition

Stress is the restoring force per unit area, measured in N/m^2 or pascal. Strain is the ratio of change in dimension to original dimension, and has no unit. The elastic limit is the maximum stress a body can bear and still return to its original shape.

Key Idea

Hooke's law — stress $\propto$ strain — holds only up to the elastic limit. Beyond it the body deforms permanently, and beyond the breaking stress it fractures.

Working Principle

Young's modulus $Y = \text{stress/strain}$ is a constant for a given material and measures its stiffness. Steel has a much larger Young's modulus than rubber, so steel is far more elastic in the physicist's sense.

Applications

- Cranes use steel ropes because steel has a very high Young's modulus and breaking stress.
- Bridge girders are designed to keep stress well below the elastic limit.
- Springs in a balance obey Hooke's law over their working range.

✓ Points to Remember (Exam Focus)

- ✓ Stress = F/A (unit N/m^2 or Pa); Strain = $\Delta L/L$ (no unit).
- ✓ Hooke's law: stress $\propto$ strain, valid only within the elastic limit.
- ✓ Young's modulus Y = stress/strain; unit N/m^2 .
- ✓ Steel is MORE elastic than rubber in physics, because it resists deformation better.
- ✓ Beyond the elastic limit the body suffers permanent (plastic) deformation.

⚠ Common Student Mistakes

- ⚠ Saying rubber is more elastic than steel — in physics elasticity means resistance to deformation.
- ⚠ Giving strain a unit; it is a pure ratio.
- ⚠ Applying Hooke's law beyond the elastic limit, where it does not hold.



Memory Hook

'Stress causes Strain; Y measures how Stubborn the material is.'



Advanced Insights

The area under a stress–strain graph gives the energy stored per unit volume. Rubber stores far more energy per unit volume than steel despite its low modulus, which is why catapults are made of rubber and not steel.

Formula Section



Stress

Difficulty	● Easy ★☆☆
Formula	$\text{Stress} = F / A$
Meaning of Variables	F = applied force in N, A = cross-sectional area in m^2 .
When to Use	To find the internal force per unit area in a stretched wire, rod or cable.
Common Mistakes	Using the surface area instead of the cross-sectional area.
Shortcut / Memory Trick	Stress has the same units as pressure.



Young's Modulus

Difficulty	● Medium ★★☆☆
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Formula	$Y = (F L) / (A \Delta L)$
Meaning of Variables	Y = Young's modulus in N/m^2 , F = force, L = original length, A = cross-sectional area, ΔL = extension.
When to Use	To find the extension of a wire under load, or to compare the stiffness of materials.
Common Mistakes	Interchanging L and ΔL in the formula.
Shortcut / Memory Trick	Y = stress $\div$ strain — build it from the two simpler definitions.

8.2 Pressure in Fluids and Pascal's Law

🕒 **Estimated time:** 14 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Pressure at a point in a fluid acts equally in all directions and increases with depth; Pascal's law states that pressure applied to an enclosed fluid is transmitted undiminished throughout it.

Real-Life Example

A hydraulic car lift raises a tonne of vehicle with a small push on a narrow piston — Pascal's law multiplying force by area ratio.

Detailed Explanation

Definition

Pascal's law states that pressure applied to an enclosed fluid is transmitted equally and undiminished in all directions throughout the fluid. Atmospheric pressure is the pressure exerted by the weight of the air column above.

Key Idea

Because pressure is transmitted undiminished, a small force on a small piston produces a large force on a large piston. Force is multiplied by the ratio of areas — but the small piston must move much further, so no energy is created.

Working Principle

Pressure at depth h is $P = h\rho g$, independent of the shape or cross-section of the vessel. This is why a dam is built thicker at the base and why water finds its own level.

Applications

- Hydraulic brakes, hydraulic press and hydraulic lift all use Pascal's law.
- Atmospheric pressure at sea level is about $1.013 \times 10^5 \text{ Pa}$, equal to 76 cm of mercury.
- A barometer measures atmospheric pressure; a manometer measures gas pressure.

✓ Points to Remember (Exam Focus)

- ✓ Pressure at depth: $P = h\rho g$; it depends only on depth, density and g .
- ✓ Pressure acts equally in all directions at a point in a fluid.
- ✓ Pascal's law: pressure in an enclosed fluid is transmitted undiminished.
- ✓ Hydraulic machines multiply force by the ratio of piston areas.
- ✓ Atmospheric pressure $\approx 10^5 \text{ Pa} \approx 76 \text{ cm}$ of mercury at sea level.

⚠ Common Student Mistakes

- ⚠ Thinking pressure at depth depends on the volume or shape of the container; it does not.
- ⚠ Believing a hydraulic press creates energy — it multiplies force, not work.
- ⚠ Forgetting to add atmospheric pressure when absolute pressure is required.



Memory Hook

'Same pressure everywhere, bigger area, bigger force' — the hydraulic bargain, paid for in distance.



Advanced Insights

The hydraulic press obeys conservation of energy exactly: the small piston moves a large distance and the large piston a small one, so $F_1 d_1 = F_2 d_2$. Force multiplication always costs distance.

Formula Section



Pressure at Depth

Difficulty	● Easy ★☆☆
Formula	$P = h \rho g$
Meaning of Variables	P = pressure in Pa, h = depth in m, ρ = density in kg/m^3 , $g = 9.8 \text{ m/s}^2$.
When to Use	To find pressure at any depth in a liquid — dams, tanks, submarines.
Common Mistakes	Including the container's area, which has no effect on pressure at depth.
Shortcut / Memory Trick	Shape does not matter — only vertical depth does.



Hydraulic Press

Difficulty	● Medium ★★☆☆
Formula	$F_1/A_1 = F_2/A_2$

Meaning of Variables	F_1 , A_1 = force and area of the small piston; F_2 , A_2 = force and area of the large piston.
When to Use	For hydraulic lifts, presses and brake systems.
Common Mistakes	Using the radius instead of the area — remember area varies as the square of the radius.
Shortcut / Memory Trick	Force multiplies by A_2/A_1 ; doubling the radius quadruples the force.

II Quick Recap — Topics Covered So Far

- Elasticity, Stress and Strain — Elasticity is the property by which a body regains its original shape after the deforming force is removed, and Hooke's law states that stress is proportional to strain within the elastic limit.
- Pressure in Fluids and Pascal's Law — Pressure at a point in a fluid acts equally in all directions and increases with depth; Pascal's law states that pressure applied to an enclosed fluid is transmitted undiminished throughout it.

8.3 Surface Tension, Viscosity and Fluid Flow

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔑 Quick Summary

Surface tension makes a liquid surface behave like a stretched membrane, viscosity is the internal friction opposing flow, and streamline flow becomes turbulent above a critical speed.

🌐 Real-Life Example

A needle floats on water despite being denser, honey pours far more slowly than water, and smoke from an incense stick rises smoothly before breaking into swirls.

📖 Detailed Explanation

Definition

Surface tension is the force per unit length acting along the surface of a liquid, arising from the cohesive forces between its molecules. Viscosity is the property of a fluid by which it opposes relative motion between its layers.

Key Idea

A molecule inside a liquid is pulled equally in all directions, but one at the surface is pulled only inward and sideways. This unbalanced inward pull is what makes the surface behave like a stretched skin and minimise its area.

Working Principle

In streamline flow every particle follows the same path at a given point. Above a critical velocity the flow becomes turbulent and eddies form. Terminal velocity is the constant speed a falling body reaches when viscous drag balances its weight.

Applications

- Raindrops and soap bubbles are spherical because a sphere has the least surface area for a given volume.
- Detergents lower the surface tension of water, letting it wet and penetrate fabric.
- Lubricating oils are chosen by viscosity; viscosity of liquids falls with rising temperature.

✓ Points to Remember (Exam Focus)

- ✓ Surface tension unit: N/m ; it decreases as temperature rises.
- ✓ Liquid drops are spherical because a sphere has minimum surface area for a given volume.
- ✓ Viscosity of liquids DECREASES with temperature; viscosity of gases INCREASES.
- ✓ Terminal velocity is reached when viscous drag plus upthrust equals weight.
- ✓ Capillary rise is caused by surface tension together with adhesion to the tube wall.

⚠ Common Student Mistakes

- ⚠ Saying viscosity of all fluids falls with temperature — it rises for gases.
- ⚠ Confusing surface tension (force per unit length) with surface energy (energy per unit area), though they are numerically equal.
- ⚠ Thinking a body at terminal velocity has no forces on it; the forces are balanced, not absent.

Memory Hook

'Surface tension pulls Inward, Viscosity drags Sideways' — one shapes drops, the other slows flow.

Advanced Insights

Whether flow is streamline or turbulent is decided by the Reynolds number, a dimensionless ratio of inertial to viscous forces. Below about 2000 flow is laminar, above about 3000 turbulent — the same criterion applies to blood vessels and to oil pipelines.

II Quick Recap — Topics Covered So Far

- Surface Tension, Viscosity and Fluid Flow — Surface tension makes a liquid surface behave like a stretched membrane, viscosity is the internal friction opposing flow, and streamline flow becomes turbulent above a critical speed.

Chapter 8 — End of Chapter Revision

All Memory Hooks

- Elasticity, Stress and Strain: 'Stress causes Strain; Y measures how Stubborn the material is.'
- Pressure in Fluids and Pascal's Law: 'Same pressure everywhere, bigger area, bigger force' — the hydraulic bargain, paid for in distance.
- Surface Tension, Viscosity and Fluid Flow: 'Surface tension pulls Inward, Viscosity drags Sideways' — one shapes drops, the other slows flow.

All Common Mistakes

-  Saying rubber is more elastic than steel — in physics elasticity means resistance to deformation.
-  Giving strain a unit; it is a pure ratio.
-  Applying Hooke's law beyond the elastic limit, where it does not hold.
-  Thinking pressure at depth depends on the volume or shape of the container; it does not.
-  Believing a hydraulic press creates energy — it multiplies force, not work.
-  Forgetting to add atmospheric pressure when absolute pressure is required.
-  Saying viscosity of all fluids falls with temperature — it rises for gases.
-  Confusing surface tension (force per unit length) with surface energy (energy per unit area), though they are numerically equal.
-  Thinking a body at terminal velocity has no forces on it; the forces are balanced, not absent.

Formula Sheet

Formula	Meaning / Use
Stress: $\text{Stress} = F / A$	To find the internal force per unit area in a stretched wire, rod or cable.
Young's Modulus: $Y = (F L) / (A \Delta L)$	To find the extension of a wire under load, or to compare the stiffness of materials.
Pressure at Depth: $P = h \rho g$	To find pressure at any depth in a liquid — dams, tanks, submarines.
Hydraulic Press: $F_1/A_1 = F_2/A_2$	For hydraulic lifts, presses and brake systems.

Quick Revision Section

- Elasticity, Stress and Strain: Elasticity is the property by which a body regains its original shape after the deforming force is removed, and Hooke's law states that stress is proportional to strain within the elastic limit.
- Pressure in Fluids and Pascal's Law: Pressure at a point in a fluid acts equally in all directions and increases with depth; Pascal's law states that pressure applied to an enclosed fluid is transmitted undiminished throughout it.
- Surface Tension, Viscosity and Fluid Flow: Surface tension makes a liquid surface behave like a stretched membrane, viscosity is the internal friction opposing flow, and streamline flow becomes turbulent above a critical speed.

Chapter 9: Heat and Thermodynamics

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that matter is made of particles in constant motion.
- Difference between the Celsius and Kelvin scales.
- Concept of energy and its conservation.
- Familiarity with melting, boiling and evaporation.

9.1 Heat, Temperature and Thermal Expansion

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Heat is energy in transit due to a temperature difference, temperature measures the average kinetic energy of particles, and most substances expand on heating.

Real-Life Example

Gaps are left between railway lines and in bridge decks so that summer expansion does not buckle the track.

Detailed Explanation

Definition

Heat is the energy that flows from a body at higher temperature to one at lower temperature. Temperature is a measure of the average kinetic energy of the particles of a body. Thermal expansion is the increase in dimensions of a body on heating.

Key Idea

A bucket of warm water contains far more heat than a red-hot spark, yet the spark is at a much higher temperature. Heat depends on mass and on temperature; temperature does not depend on mass at all.

Working Principle

Expansion may be linear, areal or cubical, each with its own coefficient, and for a given material $\alpha : \beta : \gamma = 1 : 2 : 3$. Gases expand far more than liquids, and liquids more than solids.

Applications

- Bimetallic strips in thermostats bend because two metals expand by different amounts.
- Overhead electric wires are strung slack in summer to allow for contraction in winter.
- Water shows anomalous expansion — it contracts from 0 °C to 4 °C and expands above 4 °C.

✓ Points to Remember (Exam Focus)

- ✓ SI unit of heat is the joule; 1 calorie = 4.2 J.
- ✓ Heat flows from higher to lower temperature, never the reverse on its own.
- ✓ Coefficients of expansion: α (linear), β (areal), γ (cubical), with $\beta = 2\alpha$ and $\gamma = 3\alpha$.
- ✓ Water is densest at 4 °C — this anomalous expansion lets aquatic life survive under ice.
- ✓ Temperature measures average kinetic energy; heat measures total thermal energy transferred.

⚠ Common Student Mistakes

- ⚠ Using heat and temperature interchangeably — they are different physical quantities.
- ⚠ Forgetting that water contracts between 0 °C and 4 °C instead of expanding.
- ⚠ Taking $\beta = 3\alpha$ and $\gamma = 2\alpha$, which reverses the correct relations.



Memory Hook

'Heat is what flows, Temperature is what tells it which way to flow.'



Advanced Insights

Water's anomalous expansion means ice floats and lakes freeze from the top down. If water behaved normally, lakes would freeze solid from the bottom and freshwater life would be impossible in temperate climates.

Formula Section



Linear Expansion

Difficulty	● Easy ★☆☆
Formula	$\Delta L = L \alpha \Delta T$
Meaning of Variables	ΔL = increase in length, L = original length, α = coefficient of linear expansion, ΔT = rise in temperature.
When to Use	For rods, rails, bridges and wires subjected to a temperature change.
Common Mistakes	Using the final length instead of the original length.
Shortcut / Memory Trick	$\beta = 2\alpha$ and $\gamma = 3\alpha$ for the same material.

9.2 Specific Heat Capacity and Calorimetry

⌚ Estimated time: 16 mins Difficulty: ● Medium ★★☆☆ Exam Importance: ★★★★★

 Quick Summary

Specific heat capacity is the heat needed to raise the temperature of 1 kg of a substance by 1 °C, and calorimetry uses the fact that heat lost by a hot body equals heat gained by a cold one.

 Real-Life Example

Sea breezes exist because water has a much higher specific heat than sand, so land heats and cools far faster than the sea.

 Detailed Explanation**Definition**

Specific heat capacity c is the amount of heat required to raise the temperature of unit mass of a substance by one degree. Its SI unit is J/kg·K. The principle of calorimetry states that heat lost by the hot body equals heat gained by the cold body, in an insulated system.

Key Idea

Water has an unusually high specific heat capacity of 4200 J/kg·K. This is why it is the standard coolant in car radiators and why coastal climates are far milder than inland ones.

Working Principle

Heat gained or lost with a change of temperature is $Q = mc\Delta T$. Heat absorbed during a change of state is $Q = mL$, with no temperature change. A complete problem may need both, applied stage by stage.

Applications

- Car radiators and industrial cooling systems use water for its high specific heat.
- Hot-water bottles stay warm for a long time for the same reason.
- Mixing problems: heat lost by hot water = heat gained by cold water plus calorimeter.

 Points to Remember (Exam Focus)

- ✓ $Q = mc\Delta T$ for a temperature change; $Q = mL$ for a change of state.
- ✓ Specific heat of water = 4200 J/kg·K = 1 cal/g·°C — the highest among common liquids.
- ✓ Latent heat of fusion of ice = 3.34×10^5 J/kg; of vaporisation of water = 22.5×10^5 J/kg.
- ✓ Water equivalent of a calorimeter is the mass of water that would absorb the same heat.
- ✓ During a change of state the temperature stays constant even though heat is supplied.

 Common Student Mistakes

- ⚠ Using $Q = mc\Delta T$ during melting or boiling, where ΔT is zero and the formula gives zero.
- ⚠ Forgetting the calorimeter's own heat capacity in a mixing problem.
- ⚠ Mixing units — specific heat in J/kg·K with mass in grams.

 Memory Hook

'Temperature changing $\rightarrow mc\Delta T$; State changing $\rightarrow mL$ ' — never both at once.

 Advanced Insights

Water's high specific heat comes from hydrogen bonding: much of the supplied energy goes into breaking bonds rather than raising molecular speeds. The same bonding explains its high boiling point and high latent heat of vaporisation.

Formula Section **Heat and Temperature Change**

Difficulty	● Easy ★☆☆
Formula	$Q = m c \Delta T$
Meaning of Variables	Q = heat in J, m = mass in kg, c = specific heat capacity in J/kg·K, ΔT = temperature change.
When to Use	Whenever a substance changes temperature without changing state.
Common Mistakes	Applying it across a melting or boiling point, where the state also changes.
Shortcut / Memory Trick	Break the problem into stages and apply $mc\Delta T$ and mL alternately.

 Principle of Calorimetry

Difficulty	● Medium ★★☆☆
Formula	Heat lost by hot body = Heat gained by cold body
Meaning of Variables	$m_1 c_1 (T_1 - T) = m_2 c_2 (T - T_2)$, where T is the final common temperature.
When to Use	For all mixing problems in an insulated calorimeter.
Common Mistakes	Getting the temperature differences the wrong way round, giving a negative heat.
Shortcut / Memory Trick	Always write (higher temperature – final) for the hot body and (final – lower) for the cold body.

II Quick Recap — Topics Covered So Far

- Heat, Temperature and Thermal Expansion — Heat is energy in transit due to a temperature difference, temperature measures the average kinetic energy of particles, and most substances expand on heating.

- **Specific Heat Capacity and Calorimetry** — Specific heat capacity is the heat needed to raise the temperature of 1 kg of a substance by 1 °C, and calorimetry uses the fact that heat lost by a hot body equals heat gained by a cold one.

9.3 Modes of Heat Transfer

Estimated time: 14 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Heat travels by conduction through solids, convection through fluids, and radiation through empty space, needing no medium at all.

Real-Life Example

A metal spoon in hot tea heats by conduction, the room warms by convection above a heater, and the sun reaches us across empty space by radiation.

Detailed Explanation

Definition

Conduction is heat transfer through a substance without the bulk movement of its particles. Convection is heat transfer by the actual movement of the heated fluid. Radiation is heat transfer by electromagnetic waves requiring no medium.

Key Idea

Only radiation can cross a vacuum, which is why a thermos flask has a vacuum between its walls to stop conduction and convection, and silvered surfaces to reflect radiation.

Working Principle

In conduction, particles vibrate more strongly and pass energy to neighbours while staying in place; free electrons make metals far better conductors. In convection, heated fluid becomes less dense, rises, and cooler fluid flows in to take its place, setting up a current.

Applications

- Land and sea breezes, trade winds and monsoons are all convection currents.
- A thermos flask blocks all three modes at once.
- Black surfaces are the best absorbers AND the best emitters of radiation; shiny surfaces are the worst.

✓ Points to Remember (Exam Focus)

- ✓ Conduction: mainly in solids; metals are the best conductors due to free electrons.
- ✓ Convection: only in liquids and gases, because the particles must move bodily.
- ✓ Radiation: requires no medium; travels at the speed of light.
- ✓ A good absorber of radiation is also a good emitter.

- ✓ A thermos flask uses a vacuum, silvered walls and an insulating stopper to block all three modes.

⚠ Common Student Mistakes

- ⚠ Saying convection can occur in solids — the particles cannot move bodily.
- ⚠ Thinking radiation needs air; it travels best through a vacuum.
- ⚠ Assuming a shiny surface is a good emitter; it is a poor absorber and a poor emitter.

🧠 Memory Hook

'Conduction Contacts, Convection Carries, Radiation Rays across nothing.'

🚀 Advanced Insights

The Stefan–Boltzmann law says radiated power varies as the fourth power of absolute temperature. Doubling a body's absolute temperature raises its radiated energy sixteenfold, which is why a filament glows so intensely at 2500 K and not at all at room temperature.

9.4 Laws of Thermodynamics

🕒 **Estimated time:** 14 mins **Difficulty:** ● Hard ★★★ **Exam Importance:** ★★★★★

🗣 Quick Summary

The first law of thermodynamics is conservation of energy applied to heat, and the second law states that heat cannot flow spontaneously from a colder to a hotter body.

🌐 Real-Life Example

A refrigerator does move heat from cold to hot — but only because it consumes electrical energy, exactly as the second law requires.

📖 Detailed Explanation

Definition

The first law of thermodynamics states that the heat supplied to a system equals the increase in its internal energy plus the work done by it: $\Delta Q = \Delta U + \Delta W$. The second law states that heat cannot flow of its own accord from a body at lower temperature to one at higher temperature.

Key Idea

The first law says energy is conserved; the second says which direction a process will actually go. Many processes obey the first law perfectly yet never happen, and the second law explains why.

Working Principle

Internal energy is the total kinetic and potential energy of all the particles of a system and depends only on the state, not on the path taken. No heat engine can be 100% efficient, because some heat must always be rejected to a sink.

Applications

- A heat engine converts heat into work: petrol engines, diesel engines, steam turbines.
- A refrigerator is a heat engine run backwards, using external work to move heat from cold to hot.
- The zeroth law defines temperature: two bodies in thermal equilibrium with a third are in equilibrium with each other.

✓ Points to Remember (Exam Focus)

- ✓ First law: $\Delta Q = \Delta U + \Delta W$ — conservation of energy including heat.
- ✓ Second law: heat flows spontaneously only from hot to cold.
- ✓ No heat engine can be 100% efficient; some heat must go to the sink.
- ✓ Internal energy is a state function — it depends only on the current state.
- ✓ Isothermal: constant temperature; Adiabatic: no heat exchange with surroundings.

⚠ Common Student Mistakes

- ⚠ Saying a refrigerator violates the second law; it does work, so the process is not spontaneous.
- ⚠ Confusing heat (energy in transit) with internal energy (energy stored in the system).
- ⚠ Believing a perfectly insulated engine could be 100% efficient; the second law forbids it.

**Memory Hook**

'First law: you cannot win. Second law: you cannot even break even.'

**Advanced Insights**

The second law is really a statement about entropy — disorder always increases in an isolated system. It is the only fundamental law of physics that distinguishes past from future, which is why it is called the arrow of time.

Formula Section**First Law of Thermodynamics**

Difficulty	● Medium ★★☆☆
Formula	$\Delta Q = \Delta U + \Delta W$

Meaning of Variables	ΔQ = heat supplied to the system, ΔU = increase in internal energy, ΔW = work done BY the system.
When to Use	For any thermodynamic process — isothermal, adiabatic, isobaric or isochoric.
Common Mistakes	Getting the sign of ΔW wrong; it is work done BY the system, not on it.
Shortcut / Memory Trick	Adiabatic $\rightarrow \Delta Q = 0$; Isothermal $\rightarrow \Delta U = 0$ for an ideal gas.



Efficiency of a Heat Engine

Difficulty	● Medium ★★☆☆
Formula	$\eta = W / Q_1 = 1 - Q_2 / Q_1$
Meaning of Variables	η = efficiency, W = work output, Q_1 = heat absorbed from the source, Q_2 = heat rejected to the sink.
When to Use	To compare heat engines or find the work output from heat input.
Common Mistakes	Expressing efficiency greater than 1, which is impossible.
Shortcut / Memory Trick	Efficiency is always less than 1 because Q_2 can never be zero.

II Quick Recap — Topics Covered So Far

- Modes of Heat Transfer — Heat travels by conduction through solids, convection through fluids, and radiation through empty space, needing no medium at all.
- Laws of Thermodynamics — The first law of thermodynamics is conservation of energy applied to heat, and the second law states that heat cannot flow spontaneously from a colder to a hotter body.

Chapter 9 — End of Chapter Revision

All Memory Hooks

- Heat, Temperature and Thermal Expansion: 'Heat is what flows, Temperature is what tells it which way to flow.'
- Specific Heat Capacity and Calorimetry: 'Temperature changing $\rightarrow mc\Delta T$; State changing $\rightarrow mL$ ' — never both at once.
- Modes of Heat Transfer: 'Conduction Contacts, Convection Carries, Radiation Rays across nothing.'
- Laws of Thermodynamics: 'First law: you cannot win. Second law: you cannot even break even.'

All Common Mistakes

-  Using heat and temperature interchangeably — they are different physical quantities.
-  Forgetting that water contracts between 0 °C and 4 °C instead of expanding.
-  Taking $\beta = 3\alpha$ and $\gamma = 2\alpha$, which reverses the correct relations.
-  Using $Q = mc\Delta T$ during melting or boiling, where ΔT is zero and the formula gives zero.
-  Forgetting the calorimeter's own heat capacity in a mixing problem.
-  Mixing units — specific heat in J/kg·K with mass in grams.
-  Saying convection can occur in solids — the particles cannot move bodily.
-  Thinking radiation needs air; it travels best through a vacuum.
-  Assuming a shiny surface is a good emitter; it is a poor absorber and a poor emitter.
-  Saying a refrigerator violates the second law; it does work, so the process is not spontaneous.
-  Confusing heat (energy in transit) with internal energy (energy stored in the system).
-  Believing a perfectly insulated engine could be 100% efficient; the second law forbids it.

Formula Sheet

Formula	Meaning / Use
Linear Expansion: $\Delta L = L \alpha \Delta T$	For rods, rails, bridges and wires subjected to a temperature change.
Heat and Temperature Change: $Q = m c \Delta T$	Whenever a substance changes temperature without changing state.
Principle of Calorimetry: Heat lost by hot body = Heat gained by cold body	For all mixing problems in an insulated calorimeter.
First Law of Thermodynamics: $\Delta Q = \Delta U + \Delta W$	For any thermodynamic process — isothermal, adiabatic, isobaric or isochoric.

Efficiency of a Heat Engine: $\eta = W / Q_1 = 1 - Q_2/Q_1$

To compare heat engines or find the work output from heat input.

⚡ Quick Revision Section

- **Heat, Temperature and Thermal Expansion:** Heat is energy in transit due to a temperature difference, temperature measures the average kinetic energy of particles, and most substances expand on heating.
- **Specific Heat Capacity and Calorimetry:** Specific heat capacity is the heat needed to raise the temperature of 1 kg of a substance by 1 °C, and calorimetry uses the fact that heat lost by a hot body equals heat gained by a cold one.
- **Modes of Heat Transfer:** Heat travels by conduction through solids, convection through fluids, and radiation through empty space, needing no medium at all.
- **Laws of Thermodynamics:** The first law of thermodynamics is conservation of energy applied to heat, and the second law states that heat cannot flow spontaneously from a colder to a hotter body.